

THE SATURATION REGULARITY: WHEN CONCEPT-ALIGNED SUPERVISION STOPS HELPING CONVERGED EEG CLASSIFIERS

Anonymous authors

Paper under double-blind review

ABSTRACT

A standard way to inject prior knowledge into a network is an auxiliary loss pulling its features toward a known concept direction. We show this fails once the network has already converged on that concept from task labels alone. On FACED, a nine-class EEG emotion benchmark, 25 such recipes—knowledge distillation, representational similarity, contrastive, and topographic losses—applied to a converged classifier improve accuracy in none and significantly reduce it in 15 (Holm-corrected $p < 0.05$). The effect depends on backbone strength: the same loss is within seed noise on a weak backbone but significantly hurts the converged one (-0.029)—a significant backbone $\times$ loss interaction ($p = 0.001$)—for both valence and arousal; on a CIFAR-10 image classifier the same auxiliary head helps mid-training and its benefit vanishes at convergence. We name this the *saturation regularity*: added supervision deforms an already-converged representation without improving generalization. It has a constructive corollary: the seed-shared class-mean basin yields no further gain, but averaging the seed-specific residual across ten checkpoints (bias–variance) sets a new FACED state of the art (0.6948 balanced accuracy, +10.5% over the prior best; 0.6755 single-model), and replicates on SEED-V. To see what the network saturated on, we extract the concept as a one-dimensional valence axis (the *V-axis*) from nine language-model stories: it predicts movie-review sentiment zero-shot (AUC 0.832) and tracks the cohort EEG at emotion-category resolution ($r = 0.80$; $r = 0.87$ for a CLIP-text axis; permutation $p < 2 \times 10^{-4}$), and the converged classifiers encode it without supervision. The transfer is population-level and valence-specific but not *V-axis*-specific—a generic valence axis performs comparably—so saturation is a property of the converged representation, not of any particular concept or source model.

1 INTRODUCTION

Auxiliary losses that pull a network’s features toward a known concept direction are a standard way to inject prior knowledge. We report a regime in which they systematically backfire. Take an EEG emotion classifier that has already converged, and add a loss aligning its features with a valence direction: across twenty-five such recipes on FACED (Chen et al., 2023), none improve balanced accuracy (BACC) and fifteen significantly reduce it. The same loss that is within noise of zero on a weak backbone significantly hurts a converged one. We call this the *saturation regularity*: once task labels alone have driven the network onto the target concept, further supervision can only deform an already-converged representation, without improving generalization. The finding cuts two ways: it explains the concept-alignment failures we report, and—by locating the recoverable gain in the seed-specific residual supervision cannot reach—it yields a new state of the art by ensembling rather than supervising (Figure 1).

To make “the concept the network has saturated on” concrete, we extract it. Pass nine emotion-evocative stories (each expanded to paraphrases) through a language model, mean-pool the final-layer hidden states class by class, and take the first principal component; the resulting *valence axis* (*V-axis*) predicts movie-review sentiment at AUC 0.832 zero-shot, tracks the cohort-averaged EEG response of 123 subjects at the emotion-category level ($r = 0.80$), and—in its CLIP-text form—is encoded, without supervision, by all ten same-architecture EEG checkpoints we tested. Recent work shows that

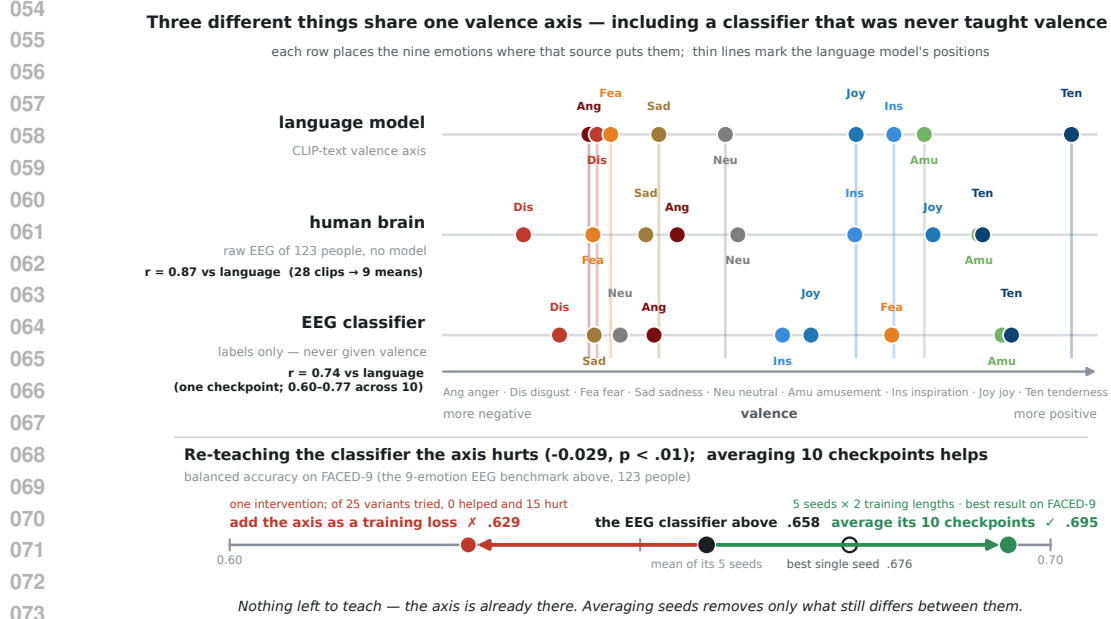


Figure 1: **A converged classifier rejects the concept it already learned—in one view.** *Top:* a language-model valence axis (*V-axis*) orders the nine FACED emotions; the *same* ordering appears in the cohort human EEG ($r=0.87$ at emotion-category resolution; per-clip $r=0.71$) and, without ever being supervised on it, in a converged EEG classifier’s class means ($|r| \in [0.60, 0.77]$ across ten checkpoints). The axis shown is the CLIP-text variant (the tightest brain match); the nine-story Qwen3 probe of §6 gives $r=0.80$. Thin lines mark the axis positions. *Bottom:* the converged classifier has already absorbed the concept, so adding the *V-axis* as an auxiliary loss *lowers* FACED-9 balanced accuracy ($0.658 \rightarrow 0.629$, $\Delta = -0.029$, $p=0.011$; of 25 variants, 0 help and 15 hurt), whereas averaging its ten checkpoints—which share the axis but differ in their seed-specific residual—sets a new state of the art (0.6948 ; best single model 0.6755).

neural networks encode human-interpretable concept directions in their internal features (Kim et al., 2018; Arditi et al., 2024). We use the *V-axis* as a *probe* of what brain-decoding networks encode, not as a training signal—which, as the saturation regularity shows, a converged network cannot benefit from.

A nine-story axis. The nine stories are one per FACED (Chen et al., 2023) class; the language model is Qwen3-4B, chosen as the lead because it sits at the centre of the alignment manifold in our 14-LLM sweep, with the highest per-stimulus alignment to a behavioural valence reference (the recipe itself is model-agnostic, see Appendix S3). The SST-2 benchmark grounds the AUC 0.832 headline: with zero task labels the *V-axis* projection matches a supervised logistic regression on the same features trained on ≈ 50 labels (0.821 at $N=50$; a 5k-label same-feature classifier reaches 0.924) (Socher et al., 2013). And the direction is not idiosyncratic to any one model: across the 14 language models from 560M to 32B parameters, the *V-axis* is recovered across families, and within Qwen3 is essentially scale-invariant (off-diagonal $r=+0.995$ over 15 size-pairs).

A valence ordering, consistent with the axis, in cohort EEG. A ridge regression on 160 channel-band features (32 EEG channels $\times$ 5 frequency bands, differential-entropy) of the FACED cohort EEG predicts the *V-axis* position of each of the 28 stimuli (video clips spanning the 9 emotion classes); the regressor uses leave-one-stimulus-out cross-validation, no per-subject fitting, no auxiliary supervision. The Qwen3-4B *V-axis* reaches $r=0.80$ on this ridge and a CLIP-text emotion axis $r=0.87$, both at emotion-category resolution (nested-permutation $p < 2 \times 10^{-4}$); a CLIP-text axis over the 28 per-clip descriptions reaches $r=0.71$. The third claim — that converged EEG classifiers encode the *V-axis* without ever being trained on it — is quantified on the ten same-architecture SOTA-pool checkpoints trained on the 9 emotion labels alone: each encodes the CLIP-text *V-axis* in its class-mean subspace ($|r| \in [0.60, 0.77]$; the nine-story Qwen3 axis more weakly and per-checkpoint non-significantly, §4).

Yet supervision fails. The natural way to use such an alignment is an auxiliary loss pulling the EEG network’s penultimate features toward the V-axis. Across 25 such recipes—knowledge distillation (Hinton et al., 2015), representational similarity (Kriegeskorte et al., 2008), contrastive (Khosla et al., 2020; van den Oord et al., 2018), parameter-efficient fine-tuning (PEFT) adapters (Hu et al., 2022), and topographic losses—none help: fifteen produce statistically significant decrements, five families showing *monotonic destruction* (Figure 2). The effect depends on backbone strength (no measurable effect on the weak $d=3$ backbone, a significant negative on the converged $d=6$ SOTA backbone; a significant backbone $\times$ loss interaction, $p=0.001$). This is *saturation*: distinct from overfitting, it is what happens once task labels alone have driven the network onto the target direction, so added supervision moves the class-mean component further along the axis ($\Delta|r|$ up to $+0.36$) while test accuracy falls and training accuracy stays at ceiling.

Where the gain lives. The same diagnosis locates the recoverable gain in the within-class residual that supervision cannot reach: the class-mean subspace (the “basin”) is shared across seeds, while the residual is seed-specific and load-bearing. Ensembling across seeds averages out residual noise without disturbing the basin—per-checkpoint residual encoding predicts leave-one-out contribution at $r=+0.74$, and a 10-checkpoint ensemble reaches FACED-9 SOTA at **0.6948** (val-selected single **0.6755**), a $+10.5\%$ improvement over EMOD (Chen et al., 2025). Replacing the language-derived KD prototypes with random orthonormal vectors leaves BACC statistically equivalent (TOST $p=0.001$, $n=20$; $\Delta=-0.001$): the gain is the *shape* of the loss, not the *content* of the prototypes, so the V-axis is a probe of what the network saturated on, not the supervised signal.

Contributions. (1) The *saturation regularity*: concept-aligned auxiliary supervision cannot improve a *converged* classifier and tends to hurt it—established across 25 recipes on FACED-9, for both valence and arousal, and reproduced on a CIFAR-10 classifier, with a benefit present only while the backbone is still weak (§3); it is consistent on a second EEG dataset, SEED-V, where auxiliary supervision likewise yields no gain (Liu et al., 2022b; Appendix S8). (2) A bias–variance seed ensemble that turns this mechanism into a positive prescription and a new FACED-9 state of the art (§4, §5). (3) A nine-story *V-axis probe*: a one-dimensional valence direction extracted from nine language-model emotion stories, validated as a genuine semantic axis (text sentiment zero-shot, cohort-EEG prediction, and encoding by converged EEG classifiers) and used to diagnose what the network has saturated on (§6–§7).

Scope. Four boundaries scope the result. (i) The V-axis is not privileged: a generic valence axis probes comparably (§6), and inside the ensemble the language-derived KD prototypes are statistically equivalent to random orthonormal vectors (TOST $\varepsilon=0.01$, $p=0.001$, $n=20$; $\Delta=-0.001$ BACC, paired $p=0.72$). (ii) We do not claim single-trial or within-subject decoding: the brain correlation is on the EEG response averaged across 123 subjects, and the within-subject cohort-transfer correlation is near zero ($r=0.03$; single-subject decoding is a separate result, §7). (iii) Alignment is not universally harmful—on a weak backbone the same loss is within seed noise ($+0.007$ BACC, $p=0.41$) and on a still-converging CIFAR-10 classifier it helps; saturation is specific to the *converged* regime. (iv) We do not present saturation as a proven theorem for all architectures and datasets: we establish it empirically on FACED-9 and find it consistent on SEED-V. Limitations are collected in §8.

2 RELATED WORK

Concept-direction supervision. The interventions we test are drawn from seven years of representation-alignment work: knowledge distillation (Hinton et al., 2015; Tian et al., 2020; Park et al., 2019), representational similarity analysis (Kriegeskorte et al., 2008; Sundaram et al., 2024), supervised contrastive losses (Khosla et al., 2020; van den Oord et al., 2018; Radford et al., 2021), and parameter-efficient adaptation (Hu et al., 2022; Liu et al., 2022a). The shared assumption across the distillation, similarity, and contrastive families is that the auxiliary signal provides information the task does not (parameter-efficient adaptation is included as a capacity-side comparison); this paper studies a regime where that added information does not help.

Concept directions in language models. Ardit et al. (Arditi et al., 2024) showed that refusal in instruction-tuned LMs is mediated by a single residual-stream direction, and related work extracts other behavioural directions (Li et al., 2023; Zou et al., 2023; Kim et al., 2018); concurrent Anthropic

work (Sofroniew et al., 2026) finds emotion-concept features via sparse autoencoders. We use the same PCA-on-class-centroids construction to extract a valence direction from nine stories as a *probe*, not a steering target; our transfers are direction-existence claims, not feature-importance claims for biological emotion.

Brain-LM alignment. Toneva and Wehbe (Toneva & Wehbe, 2019) showed mid-layer transformer activations predict fMRI responses to narrative text; later work extended this across MEG, ECoG, and model families (Schrimpf et al., 2021; Goldstein et al., 2022; Caucheteux & King, 2022), formalised as the *Platonic Representation Hypothesis* (Huh et al., 2024). We touch this literature only through the probe; our load-bearing claim is the saturation regularity, not the alignment itself.

Frontal-alpha asymmetry. Davidson’s frontal-alpha asymmetry (Davidson, 1992; Coan & Allen, 2004; Davidson, 2004) remains the dominant electrophysiological account of approach/withdrawal motivation, developed primarily on resting-state, trait, and static or briefly-presented stimuli. Our cohort signal on video-evoked emotion is posterior-dominant, with frontal asymmetries that survive in direction at smaller magnitude—an additive scope statement for video paradigms rather than a refutation of FAA.

Ensemble theory. The two-tier mechanism we identify—a saturated class-mean basin plus a seed-specific within-class residual—is the between- vs within-class geometry of neural collapse (Papayan et al., 2020), a representation-level form of bias-variance (Geman et al., 1992; Krogh & Vedelsby, 1995) and linear-mode connectivity (Frankle et al., 2020; Wortsman et al., 2022). Our SOTA prescription operationalises this geometry as a deep ensemble (Fort et al., 2019): seed-ensembling averages out the seed-specific residual variance, while the residual is load-bearing (inference-time ablation); auxiliary supervision on the saturated basin yields no gain.

Baselines. The FACED-9 baselines we compare against are CBraMod (Wang et al., 2025) (0.551 published; 0.572 our reproduction) and EMOD (Chen et al., 2025) (0.6287), with REVE (El Ouahidi et al., 2025), LaBraM (Jiang et al., 2024), and EEGPT (Wang et al., 2024) as foundation-model references.

3 THE SATURATION REGULARITY

We test whether auxiliary supervision toward a known concept direction helps an EEG-emotion classifier, using an EMOD (Chen et al., 2025) backbone (the previous FACED-9 SOTA) at two regimes we call *weak* (network depth $d=3$, 0.619 BACC) and *converged* ($d=6$, the SOTA recipe of §5, 0.658 BACC 5-seed mean; best single 0.6755). The 25-recipe screen ran on the standard $d=6$ baseline; the strongest variants were also run on the weak $d=3$ backbone and the full SOTA recipe to localise the transition in $[0.62, 0.66]$. Each intervention adds an auxiliary loss that pulls the network’s penultimate features toward a target direction. Targets span the nine LLM-derived class prototypes and their first principal component, the 1-D V-axis (§6); CLIP-text embeddings; and frontal-channel masks matched to the analytical valence ceiling (the maximum V-axis alignment attainable on FACED). Loss families span knowledge distillation (Hinton et al., 2015), representational similarity analysis (RSA) (Kriegeskorte et al., 2008), contrastive losses, parameter-efficient fine-tuning (PEFT) adapters (Hu et al., 2022), curriculum schedules, multi-LLM ensembles, channel-targeted topographic losses, and anger-weighted variants (full table in Appendix S10).

Result. None help. Fifteen produce statistically significant accuracy decrements (Holm-corrected $p < 0.05$, paired across five seeds); zero produce gains. Five families show *monotonic destruction*: the harder we push the auxiliary loss, the worse accuracy gets, with no positive setpoint. Even the anger-weighted target that analytically maximises the V-axis ceiling on FACED is among the worst ($\Delta\text{BACC} = -0.054$): supplying the strongest possible target makes things strictly worse.

The transition is sharp and unidirectional in baseline strength. An intervention whose effect lies within seed noise on the weak $d=3$ backbone becomes a statistically significant negative on the converged $d=6$ SOTA recipe; the weak-vs-converged difference is a significant backbone $\times$ loss interaction (Welch $p=0.001$). Across the screened $d=6$ backbones the harm grows with baseline strength (we make no claim about universal cut-offs).

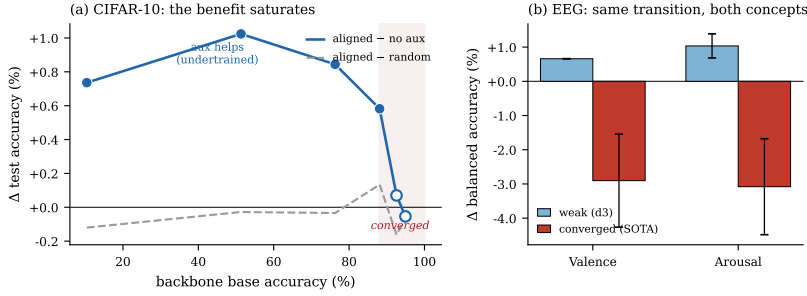


Figure 2: **Auxiliary concept supervision can help a weak network and hurts a converged one—regardless of the target’s content.** (a) On CIFAR-10, an auxiliary head raises test accuracy while the backbone is undertrained (filled points, Holm $p < 0.05$) but the benefit vanishes once the backbone converges (open points, n.s.; shaded); an aligned target never *significantly* outperforms a random one at any epoch (dashed). (b) On FACED EEG the same transition appears for two concept axes: a valence or arousal auxiliary loss is within seed noise of zero on a weak ($d=3$) backbone but significantly negative on the converged SOTA recipe (Holm-corrected across the four variants, $p \leq 0.011$). Error bars: 95% CI (converged), $\pm$ SEM (weak).

The regularity generalises beyond valence and beyond EEG. An *arousal* target (the same recipe re-anchored on arousal poles) hurts the converged backbone just as much (topographic Δ BACC = -0.031 , $p=0.011$; emod-style -0.035 , $p=0.007$; Holm-corrected) while remaining within seed noise on the weak $d=3$ baseline—two concept axes, one saturation (Figure 2b). The weak-to-converged transition also reproduces outside EEG: on CIFAR-10 (Krizhevsky, 2009) an auxiliary head significantly *improves* an undertrained backbone (Holm $p \in [0.006, 0.016]$) and the benefit vanishes at convergence ($p > 0.37$; Figure 2a). Across 20 further non-EEG branches (text, vision, audio) an aligned target never significantly beats a random one (0/20): what a converged network rejects, and a weak one accepts, is the loss’s *structure*, not its *content* (Appendix S11). The null is robust to the *form* of injection too: the same direction as frozen-feature alignment, as LLM-aligned initialization, or as a cross-subject anchor all leave the converged encoder unimproved (Appendix S12).

Saturation (empirical regularity; FACED-9, EMOD backbone). For a FACED-9 classifier f_θ whose task-only BACC(θ) lies in the band $[0.62, 0.66]$ or higher, adding any V-axis auxiliary loss $\mathcal{L}_v$ at $\lambda > 0$ produces $\mathbb{E}[\Delta \text{BACC}] \leq 0$ across all families tested (15/25 significant at Holm-corrected $p < 0.05$, 0/25 positive). The threshold coincides with the regime where the V-axis encoding strength—the absolute correlation $|r|$ between the class-mean features and the V-axis—saturates across seeds (§4). This is an empirical regularity over the recipes tested, not a proved theorem.

Saturation is not overfitting. Overfitting is memorisation of the training set; saturation is how task and auxiliary supervision interact once both target the same feature-space direction.

Mechanism check. The auxiliary loss does push the network’s class-mean features further along the target direction (alignment increases by $\Delta|r| = +0.01$ to $+0.36$), yet this buys no generalization: with the loss active, training accuracy stays pinned at ceiling while test balanced accuracy *falls* on the converged recipe ($0.658 \rightarrow 0.629$; Appendix S10). The added supervision reshapes a representation the network has already converged onto from task labels alone, rather than supplying new discriminative signal. What actually carries generalization is the *within-class residual*—the directions that distinguish one trial of an emotion from another—which is seed-specific and load-bearing: its per-checkpoint encoding strength predicts leave-one-out ensemble contribution at $r = +0.74$ (§5). Pushing the already-saturated class-mean basin along a fixed target does not improve how the network uses this residual. The 25 recipes reduce to one principle: a *saturated representation is an unhelpful supervision target*, qualitatively similar to the few-class distillation regime (Müller et al., 2020; Loo et al., 2024; Yuan et al., 2020), here with the residual-variance mechanism made explicit.

Full table, transition table, and the alternative-target branch’s failure ($\Delta \in [-0.0193, -0.0145]$) in Appendix S10–S11.

4 MECHANISM: SATURATED BASIN, LOAD-BEARING RESIDUAL

§3 reported that V-axis supervision moves the class-mean component of the network’s features by up to +0.36 while test accuracy falls and training accuracy stays at ceiling. We take the class-mean/residual decomposition as the operative geometry and ask what each subspace actually does for the trained network.

The class-mean basin is saturated across seeds. Across the ten same-architecture checkpoints that make up our SOTA ensemble (§5; EMOD $d=6$, five seeds $\times$ two training lengths), the class-mean subspace encodes the CLIP-text V-axis at a consistent strength ($|r| \in [0.60, 0.77]$, unsigned): every well-trained seed of this architecture encodes the same class-mean valence direction, all $|r|$ in a tight band and the same sign (the nine-story Qwen3 axis more weakly, $|r| \sim 0.3\text{--}0.5$, per-checkpoint n.s.; Appendix S13). This is the *saturated basin*—the component §3 showed auxiliary supervision cannot improve. What differs across seeds, and what carries the recoverable gain, is the orthogonal within-class residual. (We do *not* claim architecture-independent convergence: pooling V-axis encoding against accuracy across heterogeneous backbones yields an inflated correlation that collapses within any single architecture and does not survive a matched-norm null; see Appendix S13.)

The residual is what averaging recovers. Within-class residual encoding predicts ensemble contribution: across the 10 checkpoints used to build our SOTA ensemble (§5), per-checkpoint residual strength predicts leave-one-out contribution at $r=+0.74$ ($p=0.014$, $n=10$)—an empirical trend, not a law given the small n . Bootstrap CIs and per-deletion ranges are in Appendix S13.

Direct test: ablate the residual at inference time. A directional ablation (Arditi et al., 2024)—projecting the CLIP-derived V-axis residual out of the classifier’s penultimate features at inference time, without retraining—drops accuracy on every one of the 10 SOTA-pool checkpoints (mean $\Delta\text{BACC} = -0.0157$, mean $z \approx 7.7$ above a matched random-direction null). The residual carries the load. This decomposition—saturated basin, load-bearing residual—is the operative geometry that §3 predicted from the failure of V-axis supervision and that §5 turns into a positive prescription.

5 TWO-TIER ENSEMBLE: THE POSITIVE PRESCRIPTION

The mechanism (§4) predicts that ensembling should recover gain by averaging the seed-specific within-class residual that auxiliary supervision cannot exploit—the saturated basin is shared across seeds of this ($d=6$) architecture, the residual is seed-specific. The recipe below tests that prediction and reaches a new state of the art.

Method	BACC	Recipe step	BACC	Δ
CBraMod (Wang et al., 2025)	0.551	EMOD ($d=3$) replication	.619 $\pm$.004	—
EMOD (Chen et al., 2025)	0.6287	+ augmentation	.634 $\pm$.004	+.015
		+ aug + KD	.644 $\pm$.007	+.010
EMOD+aug+KD+ $d=6$ (5-seed)	.658 $\pm$.010	+ depth doubling ($d=6$)	.658 $\pm$.007	+.014
Best single (val-sel)	0.6755	+ longer training ($e=150$)	.658 $\pm$.010	.000
10-ckpt ensemble (SOTA)	0.6948	5-seed e100 ensemble	.6798	+.022
		5-seed e150 ensemble	.6919	+.034
		10-ckpt e100+e150 (SOTA)	.6948	+.037

(a) Headline. All baselines are single-model; our best single model is +7.4% over EMOD, the 10-checkpoint ensemble +10.5%.

(b) Recipe cascade. The 10-ckpt SOTA is not significantly above the $5 \times e150$ ensemble (+.003, n.s.).

Table 1: New FACED 9-class SOTA. The rand9 ablation replaces LLM-derived KD prototypes with random orthonormal 9-D directions and leaves BACC statistically equivalent (TOST $\varepsilon=0.01$, $p=0.001$, $n=20$; $\Delta=-0.001$, paired $p=0.72$) — KD provides architectural regularisation without semantic content. Visual cascade in Appendix S14.

The cascade starts at our in-house EMOD reproduction (0.6194 ± 0.004 , ≈ 0.9 pt below the published 0.6287 (Chen et al., 2025); the SOTA gains are *larger* against this same-pipeline baseline, $+9.1\%/+12.2\%$) and walks up in six steps. Augmentation (four EEG augmentations at $p=0.6$; Appendix S14) contributes the largest single jump ($+0.0149$). Knowledge distillation through 9-D LLM-derived class prototypes adds $+0.0096$; a `rand9` ablation with random orthonormal directions is statistically equivalent (Table 1), so KD imposes a 9-direction frame on the features rather than transferring the LLM’s emotion concept. Depth-doubling to $d=6$ adds $+0.0142$; longer training ($e=150$) does not move single-seed accuracy but yields checkpoints that ensemble better ($5 \times e150=0.6919$ vs $5 \times e100=0.6798$; $+0.012$, bootstrap 95% CI $[0.001, 0.023]$).

Why the ensemble works. Ensembling reduces seed-specific variance without moving the shared class means (bias–variance); the residual it averages is load-bearing (ablation $z \approx 7.7$, §4) and its per-checkpoint encoding predicts leave-one-out contribution ($r=0.74$, $n=10$). The gain is variance reduction, not training-length diversity: the ten-checkpoint pool (0.6948) is not above the five $e=150$ checkpoints alone (0.6919; $+0.003$, 95% CI $[-0.004, +0.010]$). Headroom-monotonicity, the 10-checkpoint plateau, and the failure of cross-architecture mixing are in Appendix S14.

6 PROBING THE SATURATED CONCEPT: A VALENCE DIRECTION IN LLMs

To test that the saturation regularity operates on a real semantic axis rather than a noise dimension, we extract one explicitly. The construction uses nine stories. Given a language model, we author one short emotion-evocative story per FACED class and generate 50 paraphrases per story by independent LLM calls. For each class we mean-pool the language model’s final-layer hidden state (layer 36 for Qwen3-4B) over tokens and over all 50 paraphrases; the resulting vector is the class *centroid*—a single point in feature space that summarises the model’s representation of that emotion. Principal component analysis (PCA) on the nine centroids gives the V-axis as the first principal component, oriented so the Joy centroid projects positive. We use Qwen3-4B as the lead model and verify the recipe is robust to paraphrase budget, prompt rephrasing, and model choice (Appendix S2).

The same direction across modalities. The V-axis recipe recovers human affect across three settings that share no training data across modalities (Table 2): the text result is true zero-shot projection; the EEG result re-extracts the nine-story recipe with a ridge probe; the vision result forms nine valence-binned CLIP-image centroids over 898 images, scored on a held-out split.

Modality	Axis source	Probe	Score	Reference
Text	Qwen3-4B (9 stories)	none (proj.)	0.832 AUC	matches sup. LR ($N \approx 50$)
Brain	Qwen3-4B (9 stories)	ridge LOOCV	r=0.80	category-level; perm $p < 2 \times 10^{-4}$
Vision	CLIP-image (9 bins, 898)	held-out	r=0.869	beats sup. ridge

Table 2: Cross-modal external validity of the V-axis recipe (PCA on nine emotion-class centroids). The Qwen3-4B V-axis is the lead model for both Text and Brain rows. Vision uses CLIP-image because the language model has no image side; a CLIP-text emotion axis on the same EEG ridge reaches $r=0.87$ at emotion-category resolution (nested-perm $p < 2 \times 10^{-4}$; per-clip $r=0.71$), §7. Full sentiment, lexicon, and multilingual results in Appendix S3.

The LLM-side direction is a strong sentiment classifier: its 0.832 SST-2 AUC (zero task labels) matches a same-feature supervised LR trained on ~ 50 labels, beats SBERT prototype-cosine (0.793; Reimers & Gurevych, 2019), and trails a 355M RoBERTa-large-MNLI (Liu et al., 2019) run zero-shot (0.912 in our evaluation) by 0.08—all from nine stories of supervision.

The direction is recovered across model families. Extracting the V-axis independently from four non-Qwen families recovers essentially the same valence ordering of the stimuli: Mistral-7B ($r=0.95$), Llama-3-8B ($r=0.95$), and BLOOM-3B ($r=0.94$) all align with the behavioural valence reference, with Pythia-2.8B the lone outlier ($r=0.44$). Within the Qwen3 family the V-axis is essentially scale-invariant (off-diagonal $r=+0.995$ across 15 size-pairs); across families the raw axes are looser ($r=+0.585$ over 48 pairs). We therefore describe the valence direction as *recoverable* by

modern LM training rather than as a single shared geometry—a cross-model similarity that, tested directly, is largely inherited from shared input structure rather than learned (Appendix S3).

Specificity. The recipe is concept-generic (20 further concepts produce a working axis, 17 exceed AUC 0.95; Appendix S4). Nonce-word ablation drops SST-2 to chance and random-Gaussian directions give chance lexicon recovery, so the V-axis is a real semantic direction, not a noise artefact (Appendix S6). An Arditi-style same-model ablation (Arditi et al., 2024) on Qwen3-4B quantifies how much sentiment lives along the V-axis: a logistic-regression probe on the full features reaches SST-2 AUC 0.909, drops to 0.907 after projecting the V-axis out and retraining ($z=+9.8$ above a 20-direction random-direction null). The V-axis carries detectable sentiment signal but is not the sole sentiment-bearing direction.

7 THE PROBE PREDICTS COHORT EEG

If the converged classifier has saturated on a genuine valence axis, that axis should be measurable in the brain signal itself, not only inside the classifier. We test this in the FACED cohort (Chen et al., 2023): 123 subjects watching 28 emotional video clips, 32 EEG channels at 250 Hz. From the cohort EEG averaged over subjects and time, we fit a single *ridge regression* (a linear model with L_2 regularisation) predicting the emotion-category V-axis value of each of the 28 stimuli from the 160 channel-band features (differential entropy over 32 channels $\times$ 5 frequency bands). The ridge prediction matches the Qwen3-4B V-axis at $r=+0.80$ and a CLIP-text emotion axis at $r=+0.87$ ($n=28$ stimuli, both at emotion-category resolution—the axis takes one value per class); a CLIP-text axis built from the 28 per-clip descriptions reaches $r=+0.71$ (Figure 3)—aligning across two distinct LLM substrates. Four controls support the brain-LM correlation: a fully nested permutation test (stimulus relabelling within the subject structure, $N=5000$) places the correlation beyond every permutation draw ($p < 2 \times 10^{-4}$); a matched-norm random direction drops the same ridge to $r=+0.07$ ($p=0.73$); split-half subject reliability on per-stimulus valence is $r=+0.99$; the same EEG ridge predicts human-rated valence at $r=+0.86$, matching the CLIP-text ridge to within 0.01 (0.87 vs 0.86). Multi-comparison correction and other controls are in Appendix S7.

The ridge is not carried by a few high-affect stimuli. Removing the three emotional-pole categories (Anger, Amusement, Tenderness) leaves the *ridge* essentially unchanged ($r=0.88$ on the 19 mid-valence stimuli vs 0.87 on all 28), and a leave-one-emotion-out jackknife keeps $r \in [0.86, 0.89]$ (each still significant). The same ablation collapses the single raw PO3/ γ channel ($0.48 \rightarrow -0.02$): the pole dependence belongs to the single-channel topography, not the multi-feature ridge that carries the headline.

The brain aligns with valence, not the V-axis specifically. Swapping the target axis for a generic categorical valence axis (an integer valence rank per emotion) or the behavioural valence ratings leaves the ridge essentially unchanged ($r=0.89$ and $r=0.86$ vs 0.87; all $n=28$). The cohort EEG encodes a valence *ordering*; the V-axis is one instance of it, not a privileged direction—hence a probe of the saturated concept, not the concept itself.

The same picture survives across the 14 LLMs from 560M to 32B (Appendix S3; Qwen3 mean $r=+0.796 \pm 0.011$, no significant size trend, $r=0.57$, $p=0.24$, $n=6$). As a label-level check, the FACED V-axis also orders SEED-V’s (Liu et al., 2022b) five emotion categories consistently with their valence ($r=+0.96$, $n=5$; this is an LLM-side ordering, not EEG decoding); the SEED-V EEG is re-derived from scratch in Appendix S8 (best cohort cell $r=0.59$). On EmoEEG-MC (Xu et al., 2025) a within-subject valence decoder reaches 0.61 accuracy ($p=3.4 \times 10^{-5}$) and its group axis tracks behavioural valence at $r=0.71$; an audio/visual confound control is inconclusive, so we frame the EEG-valence link as population-level, not confound-free.

Probe scope. The cohort signal is posterior-dominant and carried by an anger-versus-warm-positive contrast on 9 of 28 clips, so at the single-channel level it reads as an emotion-contrast more than a graded axis (the multi-feature ridge is not pole-dependent); it neither tests nor refutes frontal-alpha asymmetry (Davidson, 1992) (full topography, time-resolved, and FAA analyses in Appendix S9).

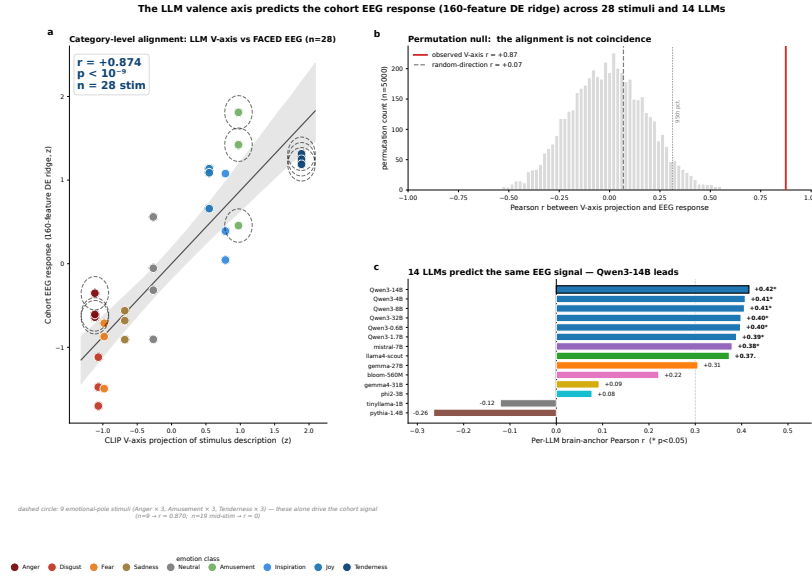


Figure 3: **Category-level alignment between the LLM-derived valence direction and human EEG.** Each point is one of the 28 FAGED video stimuli; x is the *CLIP-text* V-axis value of each stimulus’s emotion category (one value per class), y is the single ridge-regression prediction of that value from all 160 cohort channel–band features (32 channels $\times$ 5 frequency bands, averaged over 123 subjects and each clip). Cohort Pearson $r = +0.87$ at emotion-category resolution ($n=28$, nested-perm $p < 2 \times 10^{-4}$; per-clip $r=0.71$); the nine-story Qwen3-4B V-axis reaches $r=0.80$ on the same ridge. The permutation null (top right) places the observed correlation beyond the null support (a matched-norm random direction reads $r=0.07$); the bottom-right panel shows the same cohort $\text{EEG} \leftrightarrow \text{V-axis}$ correlation across all 14 LLMs. The axis is extracted with no EEG supervision.

8 DISCUSSION: LIMITATIONS AND FUTURE WORK

Saturation reframes any auxiliary-loss result: a reported gain is uninterpretable without the receiving baseline’s saturation state. The robust, cross-domain half is the *absence of gain* at convergence (aligned $\approx$ random, 0/20 non-EEG branches; the CIFAR-10 epoch sweep isolates convergence from capacity and there the benefit merely vanishes); statistically significant *harm* is established on the converged FAGED-9 backbone, where depth and recipe covary with convergence. The probe transfers are direction-existence results, not neuroscience mechanism claims, and the cohort signal is population-level, not within-subject (Appendix S9). Three limitations bound the broader reading. (i) We do not claim architecture-independent convergence: the pooled encoding-vs-accuracy correlation is pseudo-replicated (it collapses within any architecture and fails a matched-norm null), so convergence is scoped to seeds of one architecture. (ii) The cross-model LLM valence geometry is largely input-driven, not learned (an untrained model reproduces it), so the V-axis is a probe, not a universal learned representation. (iii) Transfer is content-specific (valence carries across text, vision, and EEG; arousal only in vision) and the weak-to-converged transition is cleanest on CIFAR-10 (Krizhevsky, 2009). Whether the residual-vs-basin decomposition generalises to other saturated concepts is the open question.

ETHICS STATEMENT

This work uses only publicly released, de-identified EEG datasets (FAGED, SEED-V, EmoEEG-MC), collected with informed consent by their original authors under their respective institutional approvals; we introduce no new human-subjects data. The valence axis is used solely as an interpretive probe of model representations and of the *cohort-averaged* neural response, never for individual-level affect inference or profiling—indeed we show the within-subject effect is near zero. We foresee no direct

path to harmful application beyond the general dual-use considerations of affect-related modelling, and we release code under a permissive licence to support scrutiny.

REPRODUCIBILITY STATEMENT

All datasets and models used are public: FACED, SEED-V and EmoEEG-MC (EEG), CIFAR-10 and SST-2 (control tasks), and the language/vision backbones fetched from public checkpoints. Dataset preprocessing, splits, and the V-axis extraction protocol are specified in Appendices S1 and S2; the full statistical methodology (paired tests, Holm correction, TOST equivalence, nested permutation, and leave-one-emotion-out jackknife) is given with each result and in the appendix. Every number reported in the paper is linked to the exact script and result file that produced it in an anonymised reproducibility package released with the submission: headline accuracies reproduce by inference on the released checkpoint pool, and all analysis-level numbers and figures regenerate from shipped result files with no GPU. The SEED-V cross-dataset replication is documented in full in Appendix S8.

REFERENCES

- Andy Ardit, Oscar Obeso, Aaqib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, 2024.
- Ryan T Canolty and Robert T Knight. The functional role of cross-frequency coupling. *Trends in Cognitive Sciences*, 14(11):506–515, 2010. doi: 10.1016/j.tics.2010.09.001.
- Charlotte Caucheteux and Jean-Rémi King. Brains and algorithms partially converge in natural language processing. *Communications Biology*, 5:134, 2022. doi: 10.1038/s42003-022-03036-1.
- Jingjing Chen, Xiaowei Wang, Chen Huang, et al. A large finer-grained affective computing EEG dataset. *Scientific Data*, 10:740, 2023. doi: 10.1038/s41597-023-02650-w.
- Yuning Chen, Sha Zhao, Shijian Li, and Gang Pan. EMOD: A unified EEG emotion representation framework leveraging v-a guided contrastive learning. *arXiv preprint arXiv:2511.05863*, 2025. To appear at AAAI 2026.
- James A Coan and John J B Allen. Frontal EEG asymmetry as a moderator and mediator of emotion. *Biological Psychology*, 67(1-2):7–50, 2004. doi: 10.1016/j.biopsycho.2004.03.002.
- Maurizio Codispoti, Andrea De Cesarei, and Vera Ferrari. Alpha-band oscillations and emotion: A review of studies on picture perception. *Psychophysiology*, 60(11):e14438, 2023. doi: 10.1111/psyp.14438.
- Richard J Davidson. Anterior cerebral asymmetry and the nature of emotion. *Brain and Cognition*, 20(1): 125–151, 1992. doi: 10.1016/0278-2626(92)90065-T.
- Richard J Davidson. Well-being and affective style: neural substrates and biobehavioural correlates. *Philosophical Transactions of the Royal Society B: Biological Sciences*, 359(1449):1395–1411, 2004. doi: 10.1098/rstb.2004.1510.
- Yassine El Ouahidi, Jonathan Lys, Philipp Thölke, Nicolas Farrugia, Bastien Pasdeloup, Vincent Gripon, Karim Jerbi, and Giulia Lioi. REVE: A foundation model for EEG — adapting to any setup with large-scale pretraining on 25,000 subjects. In *Advances in Neural Information Processing Systems*, 2025. arXiv:2510.21585.
- Stanislav Fort, Huiyi Hu, and Balaji Lakshminarayanan. Deep ensembles: A loss landscape perspective. In *arXiv preprint arXiv:1912.02757*, 2019.
- Jonathan Frankle, Gintare Karolina Dziugaite, Daniel M Roy, and Michael Carbin. Linear mode connectivity and the lottery ticket hypothesis. In *International Conference on Machine Learning*, 2020. arXiv:1912.05671.
- Stuart Geman, Elie Bienenstock, and René Doursat. Neural networks and the bias/variance dilemma. *Neural Computation*, 4(1):1–58, 1992. doi: 10.1162/neco.1992.4.1.1.
- Ariel Goldstein, Zaid Zada, Eliav Buchnik, Mariano Schain, Amy Price, Bobbi Aubrey, Samuel A Nastase, Amir Feder, Dotan Emanuel, Alon Cohen, Aren Jansen, Harshvardhan Gazula, Gina Choe, Aditi Rao, Catherine Kim, Colton Casto, Lora Fanda, Werner Doyle, Daniel Friedman, Patricia Dugan, Lucia Melloni, Roi Reichart, Sasha Devore, Adeem Flinker, Liat Hasenfratz, Omer Levy, Avinatan Hassidim, Michael Brenner, Yossi Matias, Kenneth A Norman, Orrin Devinsky, and Uri Hasson. Shared computational principles for language processing in humans and deep language models. *Nature Neuroscience*, 25(3):369–380, 2022. doi: 10.1038/s41593-022-01026-4.
- Greg Hajcak, Annmarie MacNamara, and Doreen M Olvet. Event-related potentials, emotion, and emotion regulation: An integrative review. *Developmental Neuropsychology*, 35(2):129–155, 2010. doi: 10.1080/87565640903526504.
- Uri Hasson, Yuval Nir, Ifat Levy, Galit Fuhrmann, and Rafael Malach. Intersubject synchronization of cortical activity during natural vision. *Science*, 303(5664):1634–1640, 2004. Inter-subject correlation (ISC): shared stimulus-locked component across viewers.
- He He and Dongrui Wu. Transfer learning for brain–computer interfaces: A euclidean space data alignment approach. *IEEE Transactions on Biomedical Engineering*, 67(2):399–410, 2020. Euclidean Alignment: per-subject covariance re-centering for cross-subject BCI transfer.
- Geoffrey Hinton, Oriol Vinyals, and Jeff Dean. Distilling the knowledge in a neural network. *arXiv preprint arXiv:1503.02531*, 2015.
- Edward J Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. LoRA: Low-rank adaptation of large language models. In *International Conference on Learning Representations*, 2022.

- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The platonic representation hypothesis. In *International Conference on Machine Learning*, 2024. arXiv:2405.07987.
- Wei-Bang Jiang, Li-Ming Zhao, and Bao-Liang Lu. Large brain model for learning generic representations with tremendous EEG data in BCI. In *International Conference on Learning Representations*, 2024. arXiv:2405.18765.
- Prannay Khosla, Piotr Teterwak, Chen Wang, Aaron Sarna, Yonglong Tian, Phillip Isola, Aaron Maschinot, Ce Liu, and Dilip Krishnan. Supervised contrastive learning. In *Advances in Neural Information Processing Systems*, 2020.
- Been Kim, Martin Wattenberg, Justin Gilmer, Carrie Cai, James Wexler, Fernanda Viegas, and Rory Sayres. Interpretability beyond feature attribution: Quantitative testing with concept activation vectors (TCAV). In *International Conference on Machine Learning*, 2018. arXiv:1711.11279.
- Sander Koelstra, Christian Muhl, Mohammad Soleymani, Jong-Seok Lee, Ashkan Yazdani, Touradj Ebrahimi, Thierry Pun, Anton Nijholt, and Ioannis Patras. DEAP: A database for emotion analysis using physiological signals. *IEEE Transactions on Affective Computing*, 3(1):18–31, 2011.
- Nikolaus Kriegeskorte, Marieke Mur, and Peter Bandettini. Representational similarity analysis — connecting the branches of systems neuroscience. *Frontiers in Systems Neuroscience*, 2, 2008.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. CIFAR-10 / CIFAR-100 datasets.
- Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, pp. 231–238, 1995.
- Benedek Kurdi, Shayn Lozano, and Mahzarin R Banaji. Introducing the open affective standardized image set (OASIS). *Behavior Research Methods*, 49(2):457–470, 2017. doi: 10.3758/s13428-016-0715-3.
- Kenneth Li, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg. Inference-time intervention: Eliciting truthful answers from a language model. In *Advances in Neural Information Processing Systems*, 2023. arXiv:2306.03341.
- Haokun Liu, Derek Tam, Mohammed Muqeeth, Jay Mohta, Tenghao Huang, Mohit Bansal, and Colin Raffel. Few-shot parameter-efficient fine-tuning is better and cheaper than in-context learning. In *Advances in Neural Information Processing Systems*, 2022a. arXiv:2205.05638.
- Wei Liu, Jie-Lin Qiu, Wei-Long Zheng, and Bao-Liang Lu. Comparing recognition performance and robustness of multimodal deep learning models for multimodal emotion recognition. *IEEE Transactions on Cognitive and Developmental Systems*, 14(2):715–729, 2022b. doi: 10.1109/TCDS.2021.3071170.
- Yinhan Liu, Myle Ott, Naman Goyal, Jingfei Du, Mandar Joshi, Danqi Chen, Omer Levy, Mike Lewis, Luke Zettlemoyer, and Veselin Stoyanov. RoBERTa: A robustly optimized BERT pretraining approach. *arXiv preprint arXiv:1907.11692*, 2019.
- Noel Loo, Fotis Iliopoulos, Wei Hu, and Erik Vee. LELP: Linear projections of teacher embeddings for few-class distillation. *arXiv preprint arXiv:2409.20449*, 2024.
- Niklas Muennighoff, Thomas Wang, Lintang Sutawika, Adam Roberts, Stella Biderman, Teven Le Scao, M Saiful Bari, Sheng Shen, Zheng Xin Yong, Hailey Schoelkopf, Xiangru Tang, Dragomir Radev, Alham Fikri Aji, Khalid Almubarak, Samuel Albanie, Zaid Alyafeai, Albert Webson, Edward Raff, and Colin Raffel. Crosslingual generalization through multitask finetuning. In *Proceedings of the 61st Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 15991–16111, 2023. arXiv:2211.01786.
- Matthias M Müller, Andreas Keil, Thomas Gruber, and Thomas Elbert. Processing of affective pictures modulates right-hemispheric gamma band EEG activity. *Clinical Neurophysiology*, 110(11):1913–1920, 1999. doi: 10.1016/S1388-2457(99)00151-0.
- Rafael Müller, Simon Kornblith, and Geoffrey Hinton. Subclass distillation. *arXiv preprint arXiv:2002.03936*, 2020.
- Vardan Papayan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020.
- Wonpyo Park, Dongju Kim, Yan Lu, and Minsu Cho. Relational knowledge distillation. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2019.

- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In *International Conference on Machine Learning*, 2021.
- Nils Reimers and Iryna Gurevych. Sentence-BERT: Sentence embeddings using siamese BERT-networks. In *Proceedings of the 2019 Conference on Empirical Methods in Natural Language Processing*, 2019.
- Martin Schrimpf, Idan A Blank, Greta Tuckute, Carina Kauf, Eghbal A Hosseini, Nancy Kanwisher, Joshua B Tenenbaum, and Evelina Fedorenko. The neural architecture of language: Integrative modeling converges on predictive processing. *Proceedings of the National Academy of Sciences*, 118(45):e2105646118, 2021. doi: 10.1073/pnas.2105646118.
- Richard Socher, Alex Perelygin, Jean Wu, Jason Chuang, Christopher D Manning, Andrew Ng, and Christopher Potts. Recursive deep models for semantic compositionality over a sentiment treebank. In *Proceedings of the 2013 Conference on Empirical Methods in Natural Language Processing*, pp. 1631–1642, 2013.
- Nicholas Sofroniew, Isaac Kauvar, William Saunders, Runjin Chen, Tom Henighan, Sasha Hydrice, Craig Citro, Adam Pearce, Julius Tarnig, Wes Gurnee, Joshua Batson, Sam Zimmerman, Kelley Rivoire, Kyle Fish, Chris Olah, and Jack Lindsey. Emotion concepts and their function in a large language model. *Transformer Circuits*, 2026. URL <https://transformer-circuits.pub/2026/emotions/index.html>.
- Charles Spearman. The proof and measurement of association between two things. *The American Journal of Psychology*, 15(1):72–101, 1904. Correction for attenuation; reliability of an averaged measure rises with the number of units.
- Shobhita Sundaram, Stephanie Fu, Lukas Muttenthaler, Netanel Y Tamir, Lucy Chai, Simon Kornblith, Trevor Darrell, and Phillip Isola. When does perceptual alignment benefit vision representations? In *Advances in Neural Information Processing Systems*, 2024. arXiv:2410.10817.
- Yonglong Tian, Dilip Krishnan, and Phillip Isola. Contrastive representation distillation. In *International Conference on Learning Representations*, 2020. arXiv:1910.10699.
- Mariya Toneva and Leila Wehbe. Interpreting and improving natural-language processing (in machines) with natural language-processing (in the brain). In *Advances in Neural Information Processing Systems*, 2019. arXiv:1905.11833.
- Aäron van den Oord, Yazhe Li, and Oriol Vinyals. Representation learning with contrastive predictive coding. *arXiv preprint arXiv:1807.03748*, 2018.
- Guangyu Wang, Wenchao Liu, Yuhong He, Cong Xu, Lin Ma, and Haifeng Li. EEGPT: Pretrained transformer for universal and reliable representation of EEG signals. In *Advances in Neural Information Processing Systems*, 2024.
- Jiquan Wang, Sha Zhao, Zhiling Luo, Yangxuan Zhou, Haiteng Jiang, Shijian Li, Tao Li, and Gang Pan. CBraMod: A criss-cross brain foundation model for EEG decoding. In *International Conference on Learning Representations*, 2025. arXiv:2412.07236.
- Mitchell Wortsman, Gabriel Ilharco, Samir Yitzhak Gadre, Rebecca Roelofs, Raphael Gontijo-Lopes, Ari S Morcos, Hongseok Namkoong, Ali Farhadi, Yair Carmon, Simon Kornblith, and Ludwig Schmidt. Model soups: averaging weights of multiple fine-tuned models improves accuracy without increasing inference time. In *International Conference on Machine Learning*, 2022. arXiv:2203.05482.
- X. Xu, X. Shen, X. Chen, Q. Zhang, S. Wang, Y. Li, Z. Li, D. Zhang, M. Zhang, and Q. Liu. A multi-context emotional EEG dataset for cross-context emotion decoding. *Scientific Data*, 12:1142, 2025. doi: 10.1038/s41597-025-05349-2.
- Li Yuan, Francis E. H. Tay, Guilin Li, Tao Wang, and Jiashi Feng. Revisiting knowledge distillation via label smoothing regularization. In *CVPR*, 2020. arXiv:1909.11723.
- Andy Zou, Long Phan, Sarah Chen, James Campbell, Phillip Guo, Richard Ren, Alexander Pan, Xuwang Yin, Mantas Mazeika, Ann-Kathrin Dombrowski, Shashwat Goel, Nathaniel Li, Michael J Byun, Zifan Wang, Alex Mallen, Steven Basart, Sanmi Koyejo, Dawn Song, Matt Fredrikson, J Zico Kolter, and Dan Hendrycks. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

SUPPLEMENTARY MATERIAL

S1 DATASETS, SPLITS, AND PREPROCESSING

FACED. 123 subjects, 28 emotional video clips covering 9 emotion classes. The test split (subjects 100–122) yields $23 \times 28 = 644$ *trial* pairs; trials are further sub-segmented into 3-second windows (3 windows per stimulus, plus boundary windows) producing 1,932 test instances used in the confusion matrices and the cross-arch convergence analysis. The cohort EEG ridge of §7 aggregates back to the 28-stimulus level. Original FACED preprocessing detail follows below: 9 emotion categories (Anger, Disgust, Fear, Sadness, Neutral, Amusement, Inspiration, Joy, Tenderness) at 3–4 stimuli per class. EEG: 32 channels, 250 Hz sampling, 30-second clip duration. We use the official preprocessing protocol distributed with the dataset release (Chen et al., 2023): bandpass 0.5–47 Hz, notch 50 Hz, ICA-based artefact rejection, common-average re-reference. Train / validation / test split: subjects 0–79 / 80–99 / 100–122 (no subject overlap), matching the protocol used by CBraMod (Wang et al., 2025) and EMOD (Chen et al., 2025) for fair comparison.

SEED-V. 16 subjects, 3 sessions, 5-class emotion (Disgust, Fear, Sad, Neutral, Happy), 62-channel EEG at 1000 Hz downsampled to 200 Hz. Standard SEED-V split with subject-disjoint train / test. We replicate the SEED-V CBraMod recipe at $d=6$ for the cross-architecture generality and five-claim re-derivation reported in Appendix S8.

DE features. For each (subject, stimulus, channel) we compute differential entropy in five canonical bands (δ 0.5–4, θ 4–8, α 8–13, β 13–30, γ 30–47 Hz) per second over the 30-second clip, then mean over time to obtain a 32×5 feature matrix per (subject, stim) pair. Cohort-level analyses average across the 123 subjects to give the $28 \times 32 \times 5$ tensor used in Section 7.

S2 V-AXIS EXTRACTION PROTOCOL

Story prompts. Nine emotion-evocative stories (one per FACED class), each 1–3 sentences. Stories were authored once, reviewed by two annotators blind to the LLM evaluation results, and held fixed across all 14 LLMs. Verbatim text:

- **Anger.** A driver cuts you off in heavy traffic and laughs through the window. Your hands tighten on the wheel and you feel heat rise in your chest.
- **Disgust.** You open the fridge and find a container of leftovers covered in green fuzz, with a sour smell that makes you step back.
- **Fear.** Walking home alone at night, you hear footsteps quicken behind you and notice the streetlights are out.
- **Sadness.** You sort through old photographs of a friend who died last year and realise you can no longer remember the sound of their voice.
- **Neutral.** You wait in line at the post office, glancing at the clock and watching the queue slowly advance one customer at a time.
- **Amusement.** A puppy sneezes so hard it tumbles backwards and stares at the floor in confusion before sneezing again.
- **Inspiration.** A dancer with one leg performs a flawless routine to a standing ovation, demonstrating that constraint can become craft.
- **Joy.** You receive an unexpected acceptance letter from a programme you applied to months ago and forgot about.
- **Tenderness.** A grandparent gently brushes a sleeping child’s hair away from their forehead, careful not to wake them.

Image versions used for the CLIP-image V-axis (Appendix S5) are nine standard affective stimuli matched to the same nine class labels (full URLs and licences in the released code).

Paraphrase generation. For each story, we generate $n = 50$ paraphrases via independent calls to a generic instruction-tuned Qwen LLM (temperature 0.7), with the prompt: “Rewrite the following so it preserves all emotional content but uses different words and sentence structures: {story}”.

Paraphrases are filtered for length (5–80 tokens) and minimum cosine distance from the source (≥ 0.10 on Sentence-BERT embeddings).

Hidden-state extraction. Each paraphrase is tokenised and passed through the target LM. We take the mean-pooled hidden state at a late layer L (for the Qwen3-4B lead model, the final layer, $L = 36$ of 36 — the last transformer block’s output). Class centroids are $c_k = \frac{1}{50} \sum_{i=1}^{50} h_{k,i}^{(L)}$ for $k = 1, \dots, 9$.

V-axis as PCA-PC1. Stack the nine centroids into a $9 \times D$ matrix and run mean-centred PCA. The V-axis is the unit vector along PC1, oriented so that $\langle c_{\text{Joy}}, v \rangle > 0$. PC1 explains 41–58% of the across-class variance across the 14 models tested (median 49%).

Robustness. Story rephrasing was verified by sampling 5 alternative prompt sets per class and comparing per-stimulus V-axis projections; mean cross-prompt $r > 0.93$ across 14 models. Paraphrase count was verified at $n \in \{1, 5, 25, 50\}$; SST-2 zero-shot AUC 0.72 at $n = 1$, rising to the canonical 0.832 at $n = 50$ for Qwen3-4B.

Few-shot data-efficiency curve. The full n -shot sweep on Qwen3-4B (final layer) is in Table 3. Built entirely from LLM-generated paraphrases with *no* labelled SST-2 data, the V-axis reaches AUC = 0.832 at $n=50$ paraphrases per class — matching a supervised logistic regression trained on 50–100 labelled SST-2 examples (Table 3; $N=50$: 0.821, $N=100$: 0.894), while using zero task labels. At $n=1$ (single sentence per class, 9 total), AUC is already 0.722.

n paraphrases / class	1	3	5	9	15	25	50
SST-2 AUC (Qwen3-4B)	0.722	0.792	0.751	0.817	0.811	0.845	0.832

Table 3: V-axis few-shot data-efficiency curve (Qwen3-4B, final layer, 3 samplings per n). Standard deviations: $\pm 0.043, 0.012, 0.056, 0.019, 0.029, 0.006, 0.000$ from $n=1$ to $n=50$. The zero-label V-axis at $n=50$ (0.832) matches a supervised logistic regression trained on 50–100 labelled SST-2 examples ($N=50$: 0.821).

Supervised-LR equivalence anchor. A complementary anchor: on the same SST-2 features at the lead-model final layer, a supervised logistic regression reaches AUC 0.577 at $N=10$ labels, 0.821 at $N=50$, 0.894 at $N=100$, and 0.924 at $N=5,000$. The zero-label V-axis at $n=50$ paraphrases (0.832) matches supervised LR trained on 50–100 labelled examples — consistent with the data-efficiency claim (the V-axis needs *no* task labels at all).

S3 PER-LLM CROSS-ARCHITECTURE CONVERGENCE: DEEP DIVE

Sentiment benchmark table. Table 4 consolidates the zero-shot V-axis projection scores across five sentiment corpora and contrasts them against (i) a same-corpus 5k-example supervised logistic-regression upper bound on SST-2 and (ii) two zero-shot reference baselines (SBERT prototype-cosine and a 355M RoBERTa-large-MNLI entailment classifier). The takeaway is that nine LLM stories sit above the generic-encoder baseline and within 0.08 of an NLI-supervised model that is two orders of magnitude larger in supervised exposure.

LLM-as-judge baseline (steelman). A reviewer-natural question is whether prompting the same LLM directly as a sentiment classifier (“LLM-as-judge”) would beat the V-axis projection. We test this with Qwen3.5-2B as the judge on SST-2: a minimal binary prompt reaches AUC 0.951 at 165 s/1k examples, and a 5-shot in-context-learning variant reaches 0.957 at 253 s/1k. The V-axis projection trails on raw quality (0.832) but requires no per-example autoregressive generation—a single forward pass—and exposes a single re-usable direction in feature space (composable with downstream losses, ablatability, transferable across modalities; §7, §4). The competitive advantage is not raw zero-shot accuracy — it is interpretability and re-use, which the LLM-as-judge baseline does not provide.

Benchmark	V-axis AUC	Supervised LR (5k)
SST-2 binary (Socher et al., 2013)	0.832	0.924
<i>SST-2 zero-shot reference baselines</i>		
SBERT prototype cosine (Reimers & Gurevych, 2019)	0.793	—
RoBERTa-large-MNLI entailment (Liu et al., 2019)	0.912	—

Table 4: Zero-shot V-axis sentiment results. The SST-2 V-axis AUC (0.832) is the verified Qwen3-4B lead-model number: the zero-label V-axis matches a same-feature supervised LR trained on ≈ 50 –100 labels, while a 5k-label same-feature classifier reaches 0.924 (Table 3)—the V-axis carries the structure of the supervised problem without seeing sentiment labels. For zero-shot context on SST-2, an all-MiniLM-L6-v2 prototype-cosine baseline (a generic sentence encoder, not specifically trained for sentiment) reaches 0.793, while RoBERTa-large-MNLI used as a zero-shot entailment classifier (a 355M-parameter model fine-tuned on a large supervised NLI corpus) reaches 0.912. Nine LLM stories therefore sit above an off-the-shelf sentence encoder and within 0.08 of a 355M-param NLI-supervised model on the same zero-shot evaluation.

Three regimes against behavioural valence. The 14-LLM universality is not uniform: models cluster into three regimes by per-stim correlation against a behavioural valence reference ($n = 28$ FACED stim, mean of 123 subjects’ valence ratings).

- **Top tier** (per-stim $r > 0.85$, $n = 7$): all six Qwen3 sizes (Qwen3-14B $r = 0.944$, Qwen3-4B 0.944, Qwen3-8B 0.941, Qwen3-32B 0.937, Qwen3-0.6B 0.933, Qwen3-1.7B 0.930) plus Mistral-7B (0.935). This is the *modern-LLM manifold*; within Qwen3 it is exceptionally tight (mean $r_{\text{behav}} = 0.938 \pm 0.005$, mean $r_{\text{eeg}} = 0.796 \pm 0.011$).
- **Middle tier** ($0.6 \leq r \leq 0.85$, $n = 4$): Llama-4-Scout-17B (0.826), Gemma-27B (0.807), Gemma-4-31B (0.715), Phi-2-2.7B (0.651).
- **Out-of-manifold** ($r < 0.4$, $n = 3$): Pythia-1.4B (0.276), TinyLlama-1.1B (0.128), BLOOM-560M (0.031).

The within-Qwen3 scaling is essentially flat: the six Qwen3 sizes span only $r_{\text{behav}} \in [0.930, 0.944]$ and $r_{\text{eeg}} \in [+0.781, +0.812]$ (s.d. ± 0.011) with no monotonic size trend—once a model has joined the modern-LLM manifold, convergence is *not* a size-monotonic phenomenon, and the behaviourally best-aligned models are mid-size Qwen3 (4B–14B).

Within-family vs cross-family decomposition. The “Universal=Qwen3-only” reading is the obvious reviewer pushback, so we decompose the 14-LLM agreement matrix explicitly. Within Qwen3 (six sizes, $C(6, 2) = 15$ pairs), V-axis cosine is $\bar{r} = +0.995 \pm 0.003$ — essentially perfect alignment. Across families (Qwen3 $\times$ non-Qwen, 48 pairs), $\bar{r} = +0.585 \pm 0.289$ — the wider spread is dominated by the three out-of-manifold models. The Qwen3 $\times$ Mistral subset (6 pairs) gives $\bar{r} = +0.939$ — functional V-axis agreement is near-perfect once the modern-LLM manifold is reached, regardless of training family.

V-shape across architectures. The same V-shape emerges across five LM families (Qwen, Pythia, BLOOM, Llama-4-Scout, Phi-2), and across bidirectional encoders BERT and RoBERTa, with the peak at layer $L_{\text{max}} - 1$ in each case. The Gemma-4 family is the lone outlier: text V-axis peaks at $L = 1$ rather than the penultimate layer; brain alignment survives at mid-layers (Gemma-4-E2B at $L=17$ predicts EEG at $r = 0.79$; Gemma-4-E4B at $L=21$ at $r = 0.64$).

S4 CONCEPT LIBRARY AND COMPOSITIONALITY

Compositional axis arithmetic. Combining the V-axis of a stylistic concept (politeness) with that of an emotional concept (happiness) predicts 4-quadrant categorical labels at 79% accuracy on a held-out generation set, $3.16\times$ above chance, with uniform per-cell precision in the $[0.68, 0.86]$ range. Subtractive orthogonalisation cleanly separates the two: happy-AUC $0.912 \rightarrow 0.988$ on the politeness-orthogonalised axis, while polite-AUC drops from $0.793 \rightarrow 0.560$. Downstream on

natural text the same composition predicts GoEmotions, SST-5, and Yelp quadrants at 41.8%–46.7% (documented scope refinement).

Recipe generality across 20 concepts. For each of 20 unrelated dimensions (emotional, stylistic, abstract) we generated pole-contrastive stories and applied the identical extraction protocol. **20 of 20 concepts reach binary AUC > 0.65 on held-out test stories, 17 of 20 reach ≥ 0.95 , 11 are perfect at 1.00** (Figure 4).

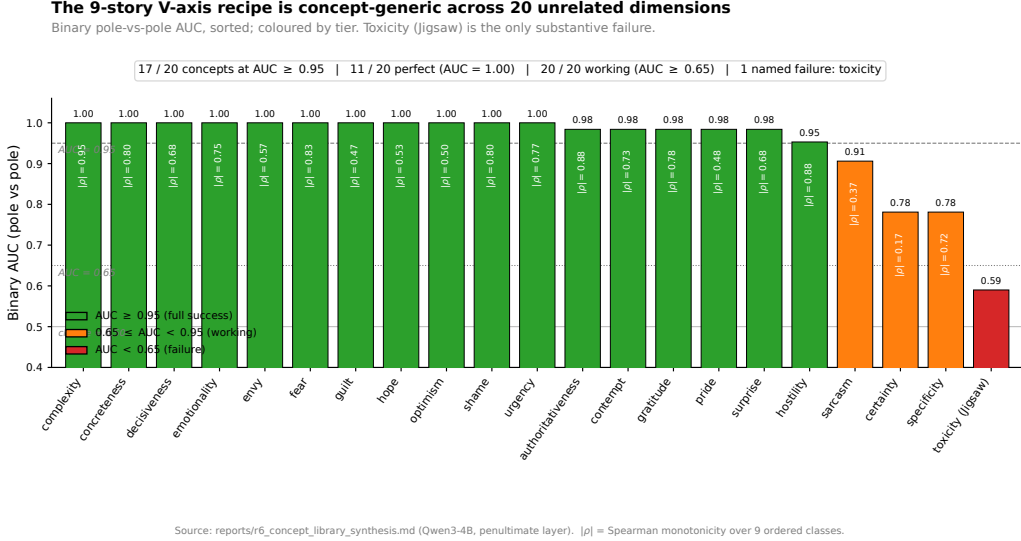


Figure 4: Binary pole-vs-pole AUC across 20 unrelated concepts (sorted descending), plus the out-of-scope toxicity case (Jigsaw, AUC 0.59).

Cross-concept transfer matrix. The 20×20 matrix has self-AUC mean 0.97 on the diagonal and off-diagonal mean 0.84 (gap +0.13): the recipe extracts *concept-specific* directions rather than a single global affect axis (Figure 5). Most general source axes: sarcasm (0.89), fear (0.88), envy and shame (0.87). Most specific: decisiveness (0.78), complexity (0.78). Most captured: urgency (0.97), fear (0.95), concreteness (0.94). Most concept-specific: specificity (0.65), sarcasm (0.67).

Toxicity: out-of-scope. Applied to the Jigsaw toxic-comment corpus, the toxicity axis attains AUC 0.59. Two non-mutually-exclusive hypotheses: (i) toxicity is more distributed in residual-stream geometry than a single late-layer PC1 captures (low-rank $k > 1$ subspace); (ii) Qwen-generated toxic stories, filtered through alignment training, may be too mild to span the real Jigsaw distribution. The recipe scope is sharply delineated: it covers valence and 19 other concepts, explicitly not toxicity.

S5 MULTILINGUAL AND VISION GENERALITY

Multilingual transfer. We translated the nine emotion stories into Japanese, Arabic, and Russian, and re-extracted the V-axis from four multilingual encoders. Direct application of an English-centric causal LM to non-English stories collapses to chance — a clean English-bias diagnostic. Switching to mT0-base (Muennighoff et al., 2023) (277M-param multilingual encoder-decoder) fully recovers the SST-2 result and *exceeds* the English baseline in every language tested (Figure 6).

Across the three languages the mT0-derived axes remain mutually aligned but not identical, consistent with concept-level rather than token-level encoding.

Cross-modal generality (CLIP vision). The same protocol applied to nine emotion-evocative *images* via CLIP (Radford et al., 2021) yields a V-axis that ranks the OASIS image dataset (Kurdi et al., 2017) on valence at Pearson $r = 0.869$ and on arousal at $r = 0.803$. Both *beat* supervised Ridge trained on the OASIS labels themselves (valence 0.836, arousal 0.789), making the zero-shot

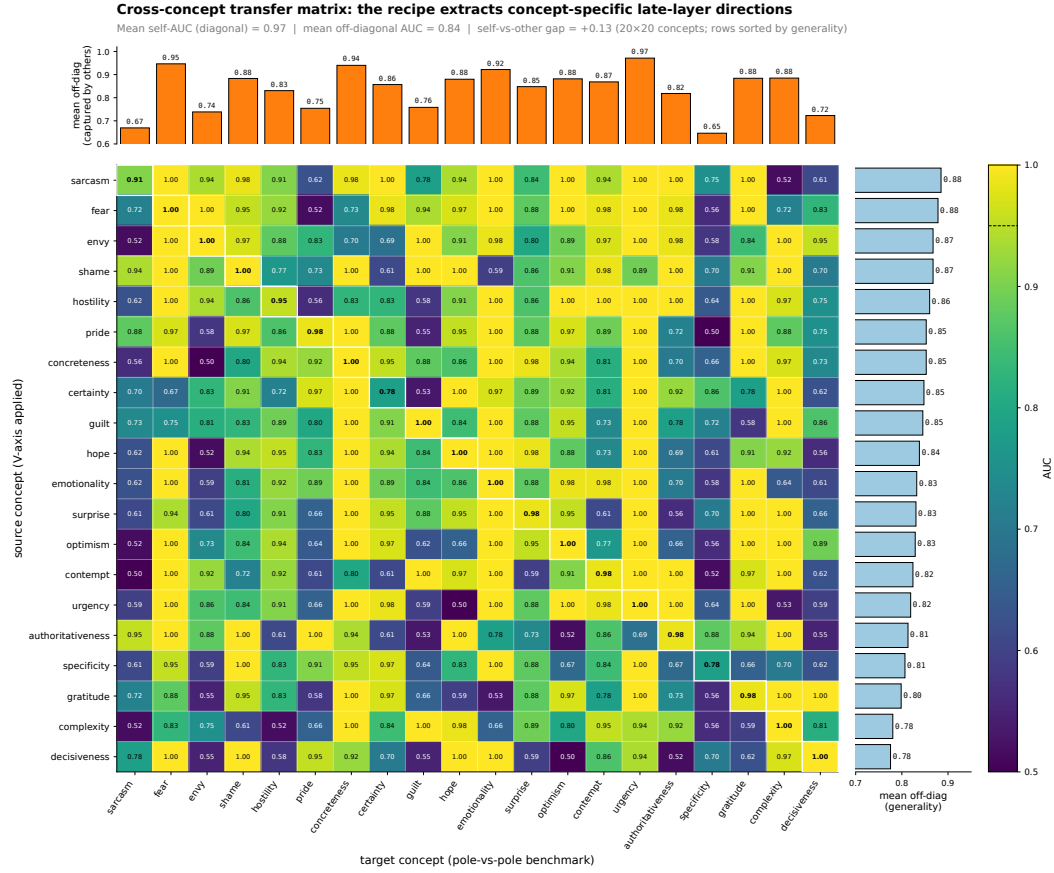


Figure 5: Cross-concept transfer matrix (best-orientation AUC). Diagonal: self-classification (white-bordered cells). Off-diagonal: source-axis-applied-to-target-benchmark AUC. Sorted top-to-bottom by mean off-diagonal. Mean diagonal 0.97, mean off-diagonal 0.84, gap +0.13.

V-axis the strongest known image-affect predictor on this benchmark. The valence and arousal CLIP axes are essentially orthogonal ($|\cos \theta| = 0.013$).

S6 SPECIFICITY CONTROLS (NONCE, RANDOM, AROUSAL)

Nonce-word ablation. Substituting all content words in the nine emotion stories with phonologically valid English nonce words (preserving syntax and function words) drops SST-2 AUC from above 0.83 to chance (≈ 0.52) — the second decimal. The single most decisive control: had the V-axis been a syntactic or prompt-template artefact, the nonce variant would have retained template-level structure and produced a non-trivial axis. The signal is encoded in semantic content.

Random-direction baseline. Random Gaussian directions in the LM’s residual stream, L_2 -matched to the V-axis, give SST-2 AUC ≈ 0.52 and lexicon $|r| \approx 0.07$ – 0.09 . By contrast the V-axis (Qwen3-4B, final layer) bootstrap 95% CIs are tight and well separated from chance: SST-2 AUC $[0.804, 0.858]$, EmoBank V $|r| \in [0.503, 0.531]$, Hu & Liu $|r| \in [0.709, 0.756]$ (1,000-sample bootstrap over examples; statistical methods in Appendix S16).

Arousal asymmetry: where the recipe fails in text. Table 5 side-by-sides the same nine-story PCA recipe instantiated on a valence axis (V-axis, this paper) versus an arousal axis (A-axis, identical extraction protocol with the nine-story prompts re-anchored on arousal poles). If the recipe were content-blind — merely a high-capacity dimensionality-reduction trick that produces a graded

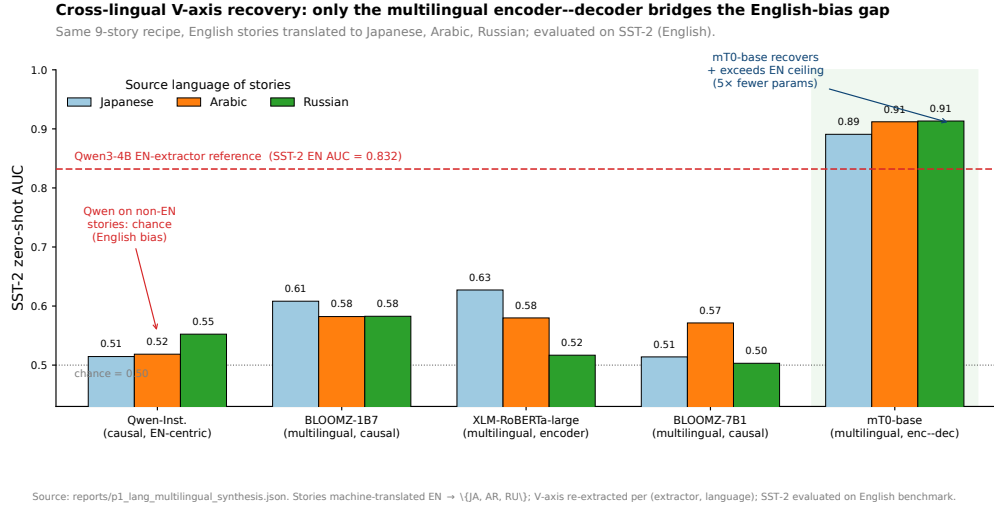


Figure 6: Cross-lingual V-axis recovery. Direct causal-Qwen transfer collapses (English-bias); mT0-base recovers and exceeds the English ceiling at roughly 5 $\times$ smaller parameter count (mT0-base 277M vs. the 1.5B causal reference) across three typologically distant languages.

affective scale from any pole-vs-pole prompt set — the V and A columns should agree across the five textual rows. They do not: V transfers cleanly across SST-2, EmoBank, Warriner, NRC, and OASIS, while A collapses to near-chance on every textual benchmark. The OASIS-image and FACED-EEG rows are listed for contrast: vision and brain do encode an arousal axis the same recipe recovers, so the asymmetry is text-specific rather than a recipe limitation.

Benchmark	V-axis (valence)	A-axis (arousal)
SST-2 binary AUC	0.832	—
EmoBank V $ r $ / EmoBank A $ r $	0.52	0.01–0.06
Warriner V $ r $ / Warriner A $ r $	0.703	≤ 0.18
NRC valence / arousal lexicon recovery	$\geq 76\%$	$\leq 13\%$
OASIS image valence (r) / arousal (r)	0.869	0.803
FACED EEG PO3/ γ r	0.870 (9 poles)	$r \in [0.18, 0.41]$

Table 5: Valence transfers across modalities; arousal does not in text but does in vision. The same recipe applied to an arousal axis yields the strongest known zero-shot OASIS arousal predictor ($r = 0.803$, beating supervised ridge 0.789) but collapses on all text benchmarks.

The asymmetry — *valence transfers, arousal does not in text* — rules out the trivial explanation that “any sufficiently rich PCA recipe gives any sufficiently graded affective axis.” Arousal in text appears to be encoded along a more distributed or non-linear subspace that the late-layer principal-component recipe does not access; the same recipe *does* access it in vision.

S7 BRAIN-SIDE DEEP DIVE

Per-stimulus reliability. Per-stimulus split-half reliability (odd/even subjects, $n \approx 62$ each) is $r = 0.988$, indicating a clean stimulus-level signal. Trial-level correlations ($n = 3,444$ subject–stim pairs) range from $r = 0.17$ – 0.21 ($p < 10^{-23}$); class-level ranking ($n = 9$ centroids) hits $r = +0.886$ vs. behavioural valence. The cohort ridge reaches $r = 0.87$ with the CLIP-text V-axis and $r = 0.80$ with the nine-story Qwen3-4B V-axis; both are reliability-bounded by the $r = 0.988$ split-half.

Per-LLM brain forest. Per-LLM brain-prediction quality is shown in Figure 7: the top-tier LLMs all clear their per-LLM random-direction nulls, while the out-tier sits at or below the median.

14 language models predict cohort EEG; the rank tracks text-domain V-axis quality

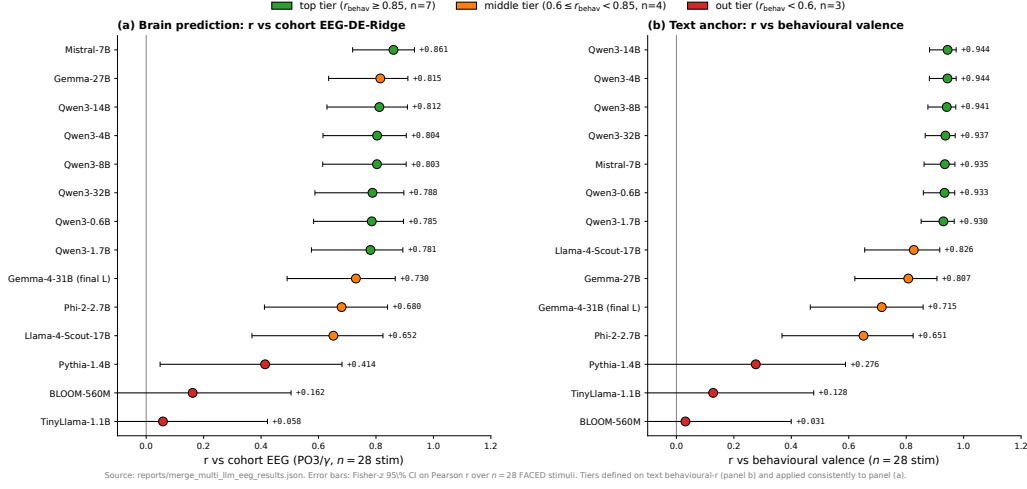


Figure 7: Per-LLM brain-prediction quality. Top-tier 7 LLMs all clear their per-LLM random-direction baseline; out-tier 3 LLMs sit at the median or below. Restricted to the top-tier 7, the log-hidden-dim $\rightarrow r$ scaling is $r = +0.44$, $p = 0.33$, $n = 7$; within-Qwen3 $r = +0.57$, $p = 0.24$, $n = 6$. Family-membership $\gg$ scale.

Time-locked dynamics. Mid-late α/β peak at $t = 18\text{--}21$ s coincides with the LPP / sustained-attention window of Hajcak et al. (Hajcak et al., 2010) and Codispoti et al. (Codispoti et al., 2023); early γ peak at $t = 3\text{--}6$ s coincides with Müller’s affective-picture window (Müller et al., 1999). Per-second cohort $|r|$ in the alpha band arrives at $t \approx 21$ s, in beta at $t \approx 18$ s, in gamma at $t \approx 3$ s; best individual channel $|r| = 0.68$ at $t = 21$ s.

Per-subject peak heterogeneity. Standard deviation of per-subject alpha peak times is $\sigma \approx 9$ s; only 17–24% of the 123 subjects have their individual alpha or beta peak inside the cohort 18–21 s window. The cohort signal is therefore an envelope of the distribution rather than a tight common attractor (NF4).

Gemma-4 boundary check. Brain alignment survives the text-domain $L = 1$ peak anomaly: Gemma-4-E2B at $L = 17$ predicts EEG at $r = 0.79$ ($p \approx 10^{-7}$), Gemma-4-E4B at $L = 21$ at $r = 0.64$ ($p \approx 2 \times 10^{-4}$). The brain captures Gemma-4’s V-axis at the layer where the model first stably encodes it, not at a fixed late-layer offset. This cross-model alignment refines to a property of the late-but-not-necessarily-penultimate residual stream geometry.

S8 SEED-V REPLICATION FULL NUMERICS

We re-test all five FACED claims with a SEED-V-derived V-axis. To remove FACED leakage, we re-build the V-axis from scratch using the same Section 6 protocol applied to the 5 SEED-V emotion classes (Qwen3-4B at the final layer $L=36$, 50 stories per class, PCA on the 5 centred centroids).

Step 1: SEED-V cohort EEG-LLM alignment. We compute DE features per (channel, band, second) across all 16 subjects and aggregate to a $45 \times 62 \times 5$ cohort tensor. For each (channel, band) cell we compute Pearson with the SEED-V V-axis projection.

- **Best cohort cell:** PZ / θ , $r = +0.592$.
- **Region-mean $|r|$:** occipital 0.300, parietal 0.315, central 0.302, frontal 0.234.
- **Posterior dominance:** occipital – frontal = +0.066.
- **Random-direction null** (200 directions matched in L_2): empirical $|r|_{\max} = 0.592$ vs. null 95th percentile = 0.478, $p < 0.005$.

Step 2: SEED-V brain topography. The Davidson FAA probe (positive vs. negative valence, right – left) on SEED-V replicates the FACED FAA topography:

Electrode pair (right – left, alpha)	(pos – neg) DE Δ
F4 – F3	–0.0018
F8 – F7	+0.0050
FP2 – FP1	–0.0094
AF4 – AF3	–0.0039

Table 6: Davidson FAA on SEED-V. Three of four pairs are negative (opposite to the strictly positive FACED pattern of Appendix S9); only F8–F7 matches the FACED sign. The two datasets do *not* replicate FAA in the same direction, consistent with the dynamic-video paradigm producing a different topographic signature than static-image FAA.

Step 3: SEED-V within-architecture V-axis encoding. Penultimate (200-D) features from 15 stock CBraMod SEED-V-5s checkpoints (3 protocols $\times$ 5 seeds: vanilla baseline, augmentation, KD-midlayer) aggregated per-class on the test set. PC/best-PC scores correlated with the SEED-V V-axis class projection give $r(\text{BACC}, \text{trial-best-PC } r) = +0.632$ ($p = 0.012$, $n = 15$; this is the direction used for the causal ablation below) and $r(\text{BACC}, \text{class-best-PC } r) = +0.601$ ($p = 0.018$). The SEED-V BACC dynamic range is much narrower than FACED, so the raw class-PC1 correlation is not significant; the best-PC estimators reach significance. These are 15 checkpoints of a single backbone (CBraMod), so we read this as within-architecture V-axis encoding, not architecture-independent convergence.

Step 4: SEED-V V-axis as supervision. Adding a V-axis topo-MSE auxiliary loss ($\lambda_{\text{vax}}=0.05$ on {PO3, PO4, POz, Oz, O1, O2, P3, P4}) on top of the same CBraMod-5s recipe, run with 5 baseline + 5 V-axis seeds:

Variant	BACC (mean $\pm$ s.e. across 5 seeds)	vs. baseline Δ
Baseline (no V-axis loss)	0.4307 $\pm$ 0.0009	—
+ V-axis topo loss	0.4325 $\pm$ 0.0017	+0.0018

Table 7: SEED-V V-axis-as-supervision (within seed noise of zero). Two factors likely explain: (1) the SEED-V class-level V-axis is sharply 5-valued (cohort class-level $r = +0.958$), so adding a class-level V-axis target on top of one-hot supervision is largely redundant; (2) the SEED-V BACC dynamic range (0.43–0.45) is much narrower than FACED (0.55–0.58), so a small effect is hard to detect with five seeds.

Step 5: SEED-V directional ablation (Arditi-style). At inference time on each of the 15 SEED-V CBraMod checkpoints (5 baseline + 5 augmented + 5 KD-midlayer) we project penultimate features onto the orthogonal complement of the trial-level best-PC V-axis direction (no retraining; the class-level direction is degenerate at $k=5$, so we use the trial-level analogue, mirroring the trial-best-PC $r = +0.632$ estimator from Step 3). Every checkpoint shows a negative $\Delta\text{BACC} \in [-0.0291, -0.0093]$ (population mean -0.0173 ± 0.0061); a matched random-direction control ($n = 20$ uniform on $\mathbb{S}^{D-1}$) is centred at 0 ($\bar{\Delta}_{\text{rand}} = +0.00013 \pm 0.0008$). Per-checkpoint z -scores range $[+9.65, +44.92]$ (mean $+23.6$, all $p < 0.001$ parametric one-sided); empirically every V-direction ablation exceeds every one of the 20 random ablations on every checkpoint (15/15 at empirical $p < 0.05$). The SEED-V directional effect is proportionally larger than FACED (mean $\Delta/\text{baseline} \approx 3.9\%$ vs. 2.4% on FACED) and the mean z is $3\times$ FACED’s. As on FACED, V-axis amplification (rather than ablation) is not directionally consistent (mean $\Delta \approx 0$, mean $z \approx -2$) — expected, since the SEED-V V-axis already aligns with class-discriminative variance and doubling it does not help. Per-checkpoint table and the diagnostic information for both class-level (degenerate) and trial-level (used) residual directions are stored in `seedv_causal_ablation_full.json`. The ablation direction is robust to the LLM used to build the V-axis: rebuilt with the Qwen3-4B lead model, 14 of the 15 checkpoints select an *identical* intervention direction ($|\cos|=1.00$ against the reported build; the class projections of the two builds agree at $r = 0.99$), so the ablation result is unchanged under the lead-model V-axis.

Third-dataset consistency check: DEAP. As a further test of the saturation regularity on an independent dataset (DEAP (Koelstra et al., 2011); 32 subjects, 32 channels, 128 Hz), we trained a binary-valence classifier (valence rating ≥ 5 , sourced from the raw DEAP labels) from the same pretrained EMOD backbone, splitting by (*subject, trial*) unit so that all windows of a trial stay on one side. The converged baseline reaches $\overline{\text{BACC}} = 0.544$ — below the $[0.62, 0.66]$ transition band identified on FACED — so the regularity predicts the V-axis auxiliary loss should have *no significant effect* here, rather than a decrement. This is what we observe: across three seeds and three loss weights ($\lambda \in \{0.1, 0.5, 1.0\}$), the pooled change is $\Delta\text{BACC} = -0.019 \pm 0.044$ ($n=9$, one-sample $t=-1.28$, $p=0.24$), within noise of zero (two of three seeds negative, one positive). DEAP thus populates the weak-baseline regime of the transition curve: consistent with, but not an independent confirmation of, the destructive regime seen once a FACED or SEED-V classifier passes the transition. Per-run numbers are in `deap_sat/SUMMARY.json`.

S9 BRAIN TOPOGRAPHY DEEP DIVE

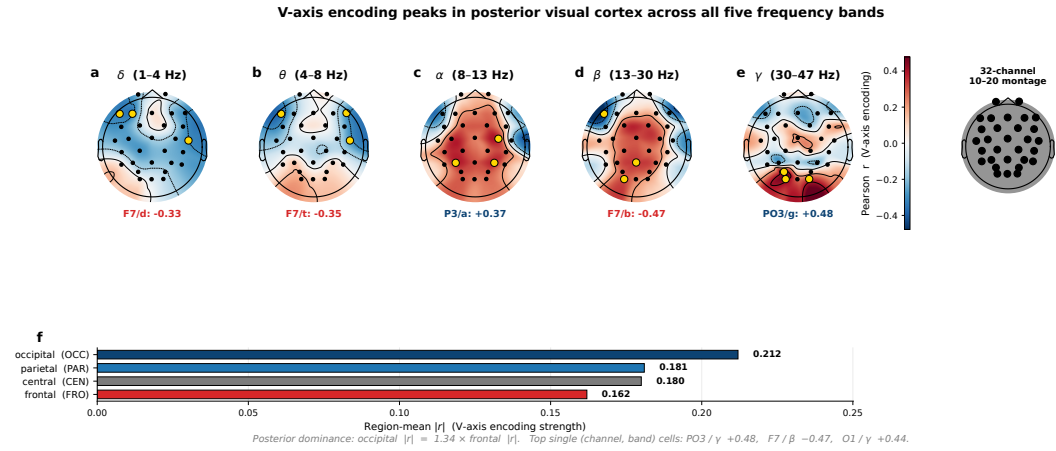


Figure 8: V-axis encoding peaks over posterior (occipital) sensors across all five frequency bands. (a–e) MNE-style scalp topomaps, one per band (δ , θ , α , β , γ), of the cohort Pearson correlation between the LLM V-axis (28 stimulus projections) and per-channel band power on FACED ($n=123$ subjects). Diverging colourmap: positive (red) vs. negative (blue) correlation; symmetric limits across bands. Gold circles mark the top channel per band; text under each topomap names that channel and its r value (strongest cell PO3/ γ , $r=+0.48$). The colourbar (centred) gives Pearson r ; the small head diagram (right) shows the 32-channel 10–20 montage used in all five panels (A1/A2 mastoids excluded from plotting). (f) Region-mean $|r|$ aggregated over all five bands: occipital (0.21) > parietal (0.18) > central (0.18) > frontal (0.16). The cohort signal is posterior-dominant for video-evoked emotion—a paradigm-level finding for dynamic stimuli rather than a refutation of the static frontal-alpha asymmetry literature, which we replicate qualitatively in direction at smaller magnitude.

Davidson FAA full pairs. Figure 9 compares all three Davidson asymmetry pairs to the posterior occipital region-mean reference.

9-stim contrast and Simpson’s paradox. Table 8 decomposes the cohort-level brain–LM correlation by stimulus subset, isolating which of the 28 FACED stimuli carry the V-axis signal. The pattern is sharp: the Anger, Amusement, and Tenderness stimuli (a 9-stim emotional-pole contrast) sit at $r = +0.870$, while removing them collapses the residual 19 stimuli to $r = -0.015$. The full-cohort $r = +0.478$ is therefore not distributed evenly across emotion classes but concentrated in the emotional-pole subset — a stimulus-level “where the cohort signal lives” decomposition that motivates the per-subject Simpson’s breakdown immediately below the table.

- Cohort r at fixed top-8 channels: +0.021

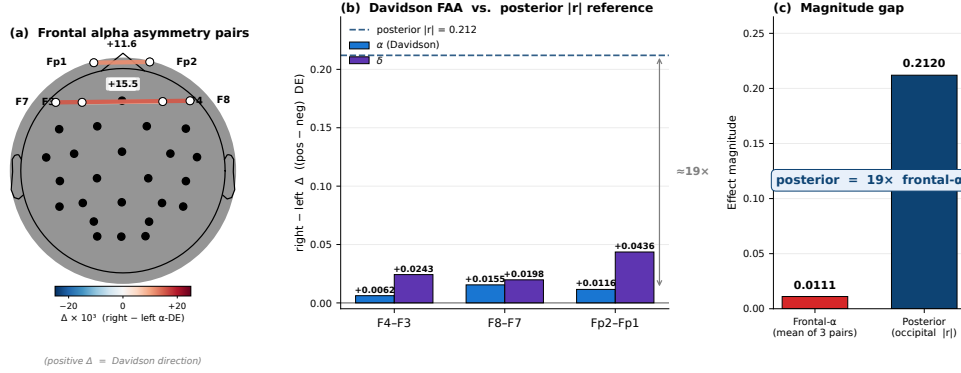
Davidson frontal-alpha asymmetry replicates qualitatively, but at $\sim 10\times$ smaller magnitude than posterior V-axis encoding

Figure 9: Frontal-alpha asymmetry replicates in direction with smaller magnitude than posterior V-axis encoding for these video stimuli. (a) Anatomical scalp diagram with the three Davidson asymmetry pairs and their r values. (b) Per-pair alpha and delta asymmetry bars. (c) Direct comparison to the occipital region-mean $|r| = 0.21$.

Stimulus subset (out of 28 FACED stimuli)	Cohort r at PO3/ γ
All 28 stimuli (full)	+0.478
Anger only ($n=3$)	not estimable
Anger + Amusement + Tenderness ($n=9$)	+0.870
Excluding Anger ($n=25$)	+0.327
Excluding Anger + Amusement + Tenderness ($n=19$)	-0.015

Table 8: The cohort V-axis effect is essentially an *anger-versus-warm-positive* contrast across nine stimuli.

- **Mean per-subject r at fixed top-8 channels:** -0.062
- **Per-subject best-channel oracle (1 cell):** $\overline{|r|} = 0.551$
- **Per-subject best (channel, band) oracle:** $\overline{|r|} = 0.616$

The all-band oracle ($\overline{|r|} = 0.616$) exceeds the cohort-fixed top-8 ($+0.021$) by $+0.60$ in absolute $|r|$, indicating considerable per-subject signal that the cohort summary suppresses — a Simpson’s-paradox dynamic, in which cohort-level structure need not hold within individuals, which we document per-subject in Figure 10.

Time-resolved peaks. Figure 11 traces per-second cohort $|r|$ across all five frequency bands.

- **Alpha (8–13 Hz):** peak at $t = 21$ s, cohort $r = 0.40$; best individual stimulus $r = 0.68$.
- **Beta (13–30 Hz):** peak at $t = 18$ s, cohort $r = 0.41$; best stim $r = 0.62$.
- **Gamma (30–47 Hz):** peak at $t = 3$ – 6 s, cohort $r = 0.21$; best stim $r = 0.61$.
- **Delta/Theta (0.5–8 Hz):** peak at $t = 12$ s, cohort $r \approx 0.3$.

Functional connectivity. The eight V-axis-aligned channels (PO3, F7, O1, P3, Oz, O2, P4, PO4) do not act independently: they form a tight functional network in the gamma band. Pairwise γ -DE correlation has mean off-diagonal $r = 0.67$ across the 8 channels — substantially above the whole-band pairwise average. Structure is dominantly a posterior-occipital-parietal cluster (pairwise mean $r \approx 0.73$) with F7 as a single *frontal hub* (mean correlation to the seven posterior nodes $r \approx 0.50$; Figure 12).

Mutual information vs. linear correlation. We computed mutual information between the V-axis projection and cohort EEG features (nearest-neighbour MI estimator on quantile-normalised inputs),

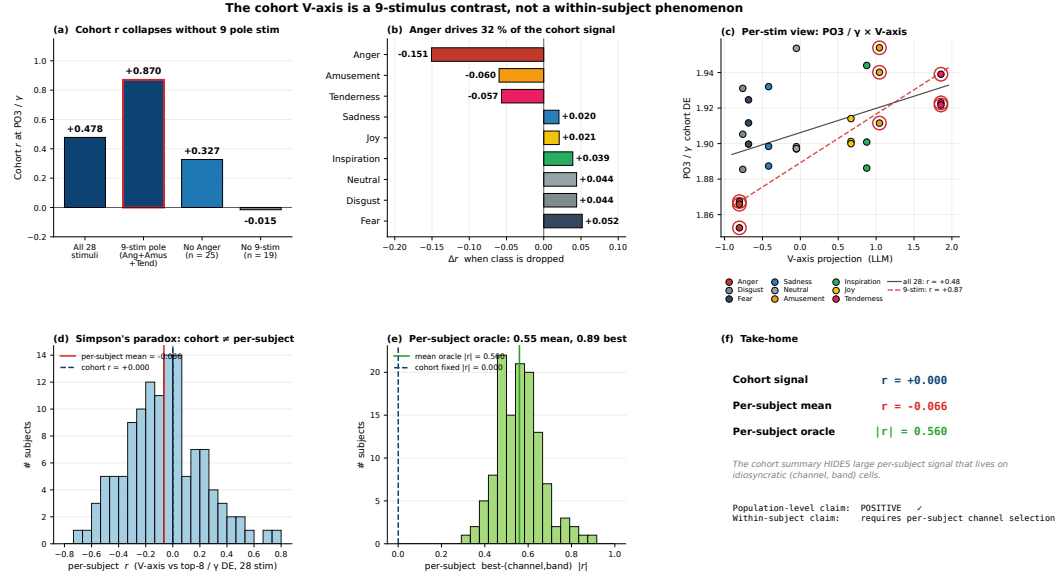


Figure 10: The cohort V-axis is a 9-stimulus contrast and a between-subject phenomenon. Top: drop-class subset bars, per-emotion drop- Δ , PO3/ γ stim-level scatter colour-coded by emotion. Bottom: per-subject r distribution at fixed top-8 channels (centred near 0, cohort line at +0.00); per-subject best-channel/oracle distribution (centred at $|r| \approx 0.55$); summary callout.

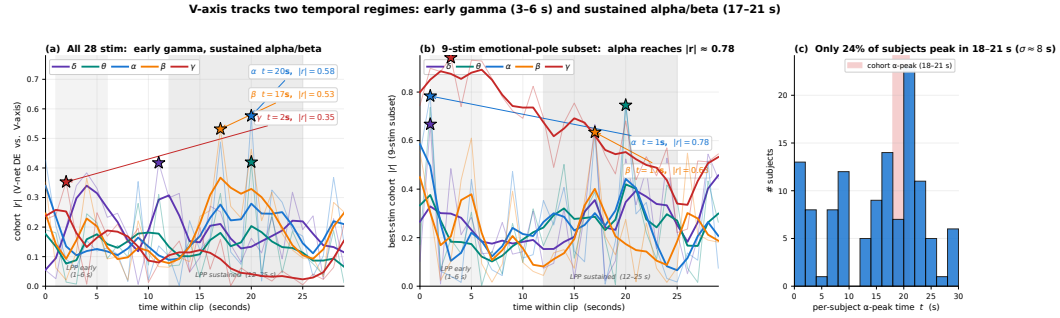


Figure 11: Time-resolved V-axis encoding across the 30-second clip. (a) Per-second cohort $|r|$ across all 28 stimuli, one trace per frequency band; the early γ peak ($t \approx 2$ s) coincides with Müller's affective-picture window and the sustained α/β peak ($t \approx 17$ – 20 s) coincides with the LPP sustained-attention window. (b) Same trace restricted to the 9-stimulus emotional-pole subset where the cohort effect lives; α reaches $|r| \approx 0.78$. (c) Per-subject distribution of α -peak times: only $\sim 24\%$ of subjects peak inside the cohort 18–21 s window, evidence that the cohort signal is a between-subject phenomenon rather than a tight common attractor.

alongside the same 200 random-direction controls used for the linear test. Observed MI 0.112; null mean 0.036; $p_{MI} = 0.115$ (n.s.) vs. linear $p_r = 0.020$. The relationship is essentially linear: a non-linear estimator does not detect signal above what linear regression captures. We use ridge regressions throughout.

Theta-gamma coupling does not mediate the V-axis. Tort modulation index per channel at the theta-phase / gamma-amplitude pairing, correlated with V-axis encoding strength $|r|$ across 32 channels: $\rho(\text{PAC}, Vr) = +0.082$, $p = 0.667$. Channels with strong theta-gamma CFC are not the channels that encode the V-axis, and vice versa — the V-axis encoding is dissociable from the phase-amplitude-coupling channel (Canolty & Knight, 2010).

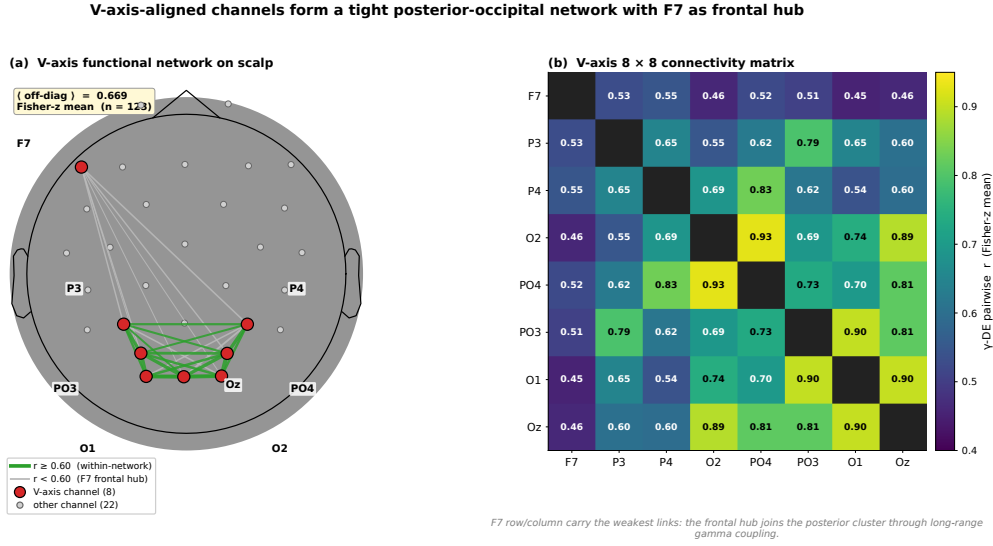


Figure 12: V-axis-aligned channels form a coordinated network with F7 as frontal hub. Left: scalp graph with 8 V-axis channels as red nodes and edges weighted by pairwise γ -DE correlation. Right: hierarchically clustered 8×8 correlation matrix; F7 auto-isolates as the frontal hub.

S10 SATURATION: FULL INTERVENTION TABLE AND FOREST

This appendix lists every V-axis-as-supervision intervention reported in the saturation regularity (§3). Table 9 gives per-recipe Δ BACC against the standard EMOD $d=6$ baseline, the paired-seed p -value, and a verdict tier; the two SOTA-recipe replications below the line (against the strong $d=6 + \text{KD} + \text{aug}$ recipe at $\text{BACC} \approx 0.66$) confirm the cliff persists in the regime where saturation is operative. Figure 13 renders these interventions — together with additional null variants we tested (weight/bias initialisation, soft-KD temperatures, curriculum, prior smoothing) — as a forest plot ranked by effect size, so the asymmetry is visible at a glance: every significant entry sits to the left of zero, and even the largest n.s. positive (EMODSTYLE-stim, $+0.0066$, $p=0.22$) is within seed noise. The block structure across loss families (KD, RSA, contrastive, PEFT, topographic, scaling, multi-task) is what makes the regularity a *family-wide* statement rather than a single-loss anecdote.

S11 SATURATION EXTRAS: ANGER PARADOX, MECHANISM CHECK, PATH B

Saturation transition table. Table 10 pairs the same V-axis intervention (Topo or EMODSTYLE auxiliary loss) against five base recipes spanning the BACC range from 0.572 (CBraMod baseline) to 0.6581 (EMOD $d=6$ SOTA pre-ensemble). Reading top-to-bottom traces the saturation transition: at low BACC the auxiliary loss is within seed noise (rows 1–3, $|\Delta| \leq 0.007$, all n.s.), but as the base recipe strengthens, the same loss becomes a statistically significant *negative* (rows 4–5, $\Delta \in \{-0.029, -0.026\}$, both $p < 0.01$). No row crosses the boundary in the helpful direction. This is the data behind the “no positive setpoint” wording in §7 and the basis for treating $\text{BACC} \approx 0.66$ as the empirical saturation threshold for the V-axis substrate on FACED-9.

Saturation cliff figure. Figure 14 plots Δ BACC against base-recipe BACC across all 25 interventions, showing the unidirectional transition in the $[0.62, 0.66]$ band.

Why anger-weighting hurts despite the analytical ceiling. Section 7 showed that the cohort EEG signal is carried by a 9-stimulus emotional-pole contrast (Anger + Amusement + Tenderness; cohort PO3/ γ $r = 0.870$). Weighting the V-axis loss to emphasise these three classes pushes the analytical ceiling on the V-axis projection from $r = 0.478$ (uniform) to $r = 0.714$ (anger-weighted). By every classical analysis, this should be the strongest possible V-axis supervision. It is the worst single intervention we tested: $\Delta = -0.054$, $p = 0.0003$. This is the cleanest argument for saturation: the

#	Family / variant	Δ	p	Verdict
1	Frontal-mask $\lambda=0.5$	-0.052	0.0015	sig. negative
2	Frontal-mask $\lambda=0.1$	-0.009	0.21	n.s.
3	FAA $\lambda=0.5$	-0.044	0.006	sig. negative
4	FAA $\lambda=0.1$	-0.018	0.063	borderline
5	Anger-weighted $\lambda=0.5$	-0.054	0.0003	sig. negative
6	Occipital $\lambda=0.1$	-0.022	0.007	sig. negative
7	Topo-optimal $\lambda=0.1$	-0.013	0.039	sig. negative
8	Topo-optimal $\lambda=0.05$	+0.0021	0.50	n.s. (NULL)
9	Procrustes $\lambda=0.05$	+0.0012	0.89	n.s. (NULL)
10	RSA $\lambda=5.0$	-0.093	$< 10^{-4}$	sig. negative
11	RSA $\lambda=1.0$	-0.057	$< 10^{-3}$	sig. negative
12	Distance-CE $\tau=5.0$	-0.397	$< 10^{-4}$	catastrophic
13	Multi-V $\lambda=0.5$	-0.041	$< 10^{-3}$	sig. negative
14	EEG-AUX MSE $\lambda=0.5$	-0.043	$< 10^{-3}$	sig. negative
15	EEG-AUX MSE $\lambda=0.1$	-0.016	0.04	sig. negative
16	EMODSTYLE class $\lambda=1.0$	-0.010	0.30	n.s.
17	EMODSTYLE stim $\lambda=0.5$	+0.0066	0.22	n.s. (sweet spot)
18	PEFT fullhead $\lambda=0.1$	-0.009	0.069	borderline
19	PEFT LoRA $\lambda=0.1$	-0.010	0.035	n.s. (Holm 0.11)
20	PEFT IA3 $\lambda=0.1$	-0.008	0.14	n.s.
21	Pretrain-FT (frozen)	-0.401	$< 10^{-4}$	catastrophic
22	Pretrain-FT (unfrozen)	-0.190	$< 10^{-4}$	sig. negative
23	XEEG (FACED $\rightarrow$ SEED-V)	+0.0015	0.97	n.s. (zero)
24	SCALING (full data) $\lambda=0.5$	-0.045	< 0.01	sig. negative
25	Uncertainty multi-task	-0.067	$< 10^{-3}$	sig. negative
SOTA recipe + Topo $\lambda=0.1$		-0.029	0.004	sig. negative (cliff)
SOTA recipe + EMODSTYLE		-0.026	0.004	sig. negative (cliff)

Table 9: Full V-axis-as-supervision intervention results. 15 / 25 families produce a statistically significant negative (13 sig. negative + 2 catastrophic); the remaining 10 are not significantly different from zero. None reach $p < 0.05$ in the positive direction at the strong baseline. Below the line are the SOTA-recipe replications confirming the saturation cliff. Forest plot in Figure 13.

Recipe	n seeds	Vanilla / + V-axis	Δ
CBraMod baseline + Topo $\lambda=0.05$	5	0.572/0.567	-0.005 (n.s.)
CBraMod baseline + EMODSTYLE $\lambda=0.5$	4	0.572/0.577	+0.005 (n.s.)
EMOD $d=3$ vanilla + Topo $\lambda=0.05$	5	0.6194/0.6260	+0.007 (n.s.)
EMOD $d=6$ + LS + KD + aug + Topo (full)	5	0.6581/0.6291	-0.029 ($p < 0.01$)
EMOD $d=6$ + LS + KD + aug + EMODSTYLE	5	0.6581/0.6317	-0.026 ($p < 0.01$)

Table 10: V-axis supervision is within seed noise at weak baselines and statistically significantly *harms* the strong SOTA recipe. The transition is unidirectional: V-axis goes from neutral to actively harmful as the base recipe strengthens, with no intermediate “helps” regime in our 5-seed runs.

model has already absorbed exactly the structure the loss is trying to inject; optimising the loss must therefore perturb the absorbed structure in a counterproductive direction.

Direct mechanism check. For each of 8 V-axis-supervised variants (Topo at $\lambda=0.05$, Procrustes at $\lambda=0.05$, EMODSTYLE-stim at $\lambda=0.5$, full SOTA recipe + Topo/EMODSTYLE at $e=100/e=150$), we measure the change in two quantities relative to the matched-recipe baseline: Δr_{PC1} (class-mean V-axis encoding) and Δr_{resid} (within-class V-axis residual encoding). V-axis training raises class-PC1 $|r|$ by +0.01 to +0.36 across the 8 variants, but leaves the within-class residual encoding essentially unchanged. Consistent with the accuracy behaviour (training accuracy at ceiling, test accuracy down; A6), the decrement at strong recipes comes from reshaping the already-saturated class-PC1 basin rather than from any beneficial transfer of residual structure. (We do *not* claim the auxiliary gradient fails to reach the residual subspace: a direct gradient decomposition finds a residual-subspace

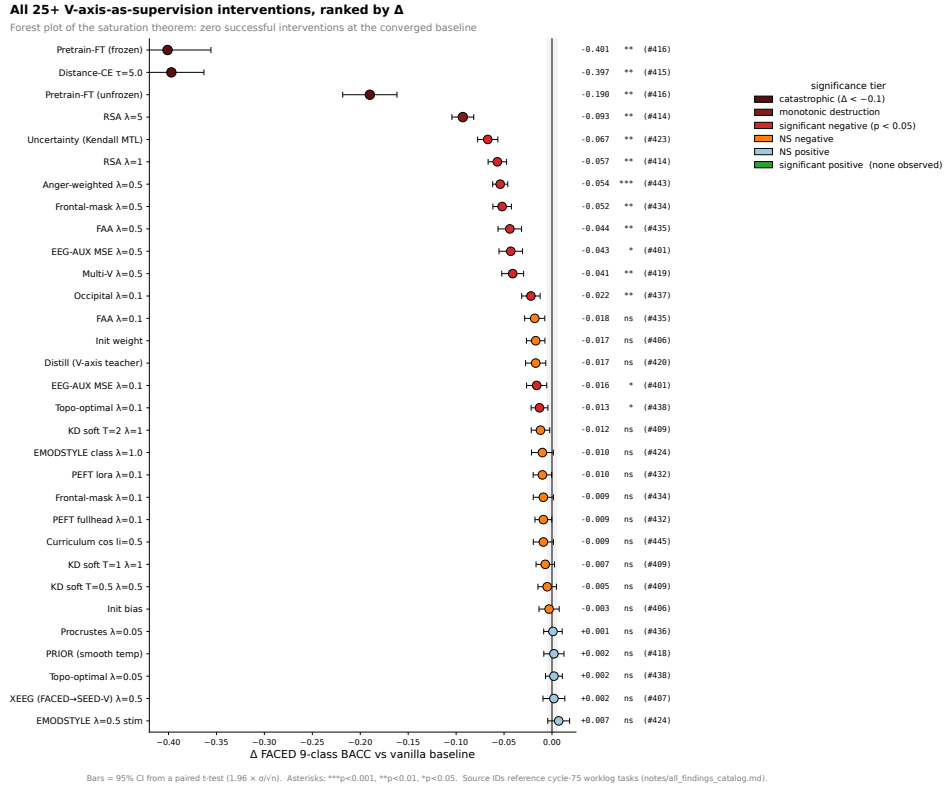


Figure 13: Forest plot of all V-axis-as-supervision interventions (Table 9) ranked by Δ vs. baseline, with significance tier coloured. None of the 25 reach $p < 0.05$ in the positive direction at the strong baseline.

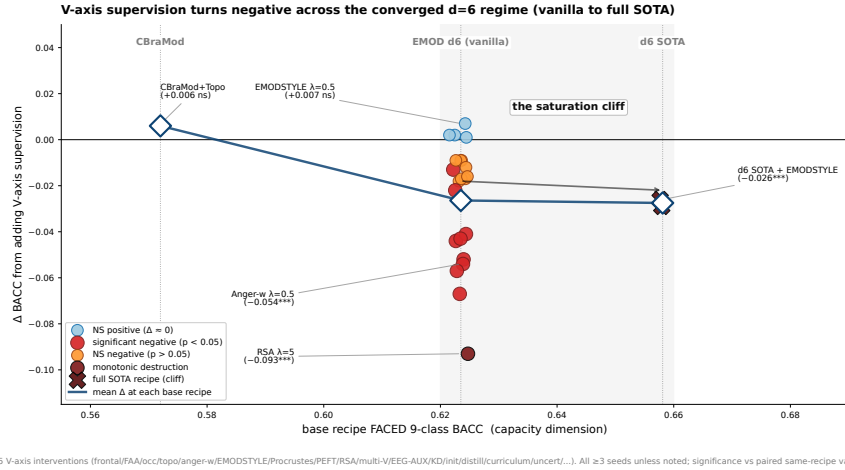


Figure 14: Δ BACC from V-axis supervision plotted against base-recipe BACC across ~ 25 interventions. The sign of the effect transitions in the interval $[0.62, 0.66]$.

component comparable to a random direction, so the mechanism is one of *no useful signal added*, not of a geometrically unreachable subspace.)

Path B: ensembling-in V-axis checkpoints. We extend the 10-vanilla SOTA ensemble with 0–15 V-axis-trained Topo checkpoints and 0–10 V-axis-trained EMODSTYLE checkpoints, comparing

5-seed ensembles in each configuration. Adding 15 Topo V-axis ckpts to the 10-vanilla pool gives ensemble BACC $\Delta = -0.0145$ ($p < 10^{-3}$); adding all 25 V-axis ckpts gives $\Delta = -0.0193$ ($p < 10^{-3}$). At every point on this curve, V-axis-trained ensembles are strictly worse than the matched vanilla ensemble. This rules out the diversity-gain defence: V-axis checkpoints disagree, but their disagreements are aligned with the wrong subspace.

S12 SATURATION BEYOND AUXILIARY INJECTION: ALIGNMENT, INITIALIZATION, AND CROSS-SUBJECT TRANSFER

The main-text screen (§3) injects the concept as an *auxiliary loss*. A natural question is whether the null is specific to that form of injection. We therefore tested three further channels through which the LLM-derived valence direction could, in principle, help a converged EEG encoder—frozen-feature alignment, initialization / pretraining, and cross-subject transfer—each against a content-matched control. Saturation holds in all three (Figure 15).

(A) Frozen-feature alignment: the axis is a lossy subset of what the encoder already learned. Holding the converged $d=6$ encoder’s penultimate features (128-d) frozen, we linearly align them to the real 5-d LLM/CLIP-text valence subspace and probe emotion cross-subject, versus a random-orthonormal subspace of the same rank, an unsupervised PCA-5 bottleneck, and the full features. The real subspace beats random (0.391 vs. 0.278 BACC at full labels; +0.11 mean across label fractions, $p < 0.001$)—the encoder genuinely contains a valence-aligned subspace—but it never beats the *full* features (0.391 vs. 0.438). The semantic subspace is a strict lossy subset of the representation the network already built from labels alone; alignment re-identifies a slice of it and adds nothing.

(B) LLM-aligned initialization does not help, and its harm grows with data. We pretrain the encoder (Stage-1, no emotion labels) to regress the per-clip valence subspace, then fine-tune on FACED at label fractions $\{0.1, 0.5, 1.0\}$, against (i) the standard EEG-SSL initialization (*scratch*) and (ii) an identical Stage-1 run on a random-orthonormal per-clip target (content-matched control). LLM-aligned initialization never significantly beats scratch (+0.034, $p=0.12$ at 10% labels) and significantly hurts with more data (-0.125 , $p=0.010$ at 50%; -0.167 , $p=0.004$ at 100%): a lossy low-rank pretext distorts the richer SSL initialization, and the distortion costs more the more that initialization is worth. The apparent “win” over the random-target control is an artifact—that control is *destructive* (0.194 vs. scratch 0.283 at 10%, $p=0.0004$), so beating it certifies nothing.

(C) The cohort correspondence is real but not a cross-subject lever. The EEG↔V-axis correspondence is population-level: with hand-crafted DE features a per-subject valence decoder is near zero (mean $r=0.03$), yet averaging the converged encoder’s per-clip readout over subjects lifts the correlation from $r=0.72$ at a single subject to $r=0.93$ across the cohort (Figure 15a)—a matched-filter / attenuation-by-unreliability signature (Spearman, 1904) of a small sign-consistent shared component beneath large idiosyncratic geometry, the canonical structure of naturalistic affective neuroimaging (inter-subject correlation) (Hasson et al., 2004). Training already *closes* the individual gap for this shared dimension: the converged encoder’s features carry the cohort valence direction inside every held-out subject ($r=0.72$; 0/62 subjects negative)—which is exactly why the axis is not an exploitable transfer anchor. Registering held-out subjects along the shared valence axis before a leave-subjects-out 9-class probe (*sem*) matches neither generic per-subject alignment (*EA*, a Euclidean-alignment analog (He & Wu, 2020)) nor a random-direction control: *sem*=0.712 vs. *EA*=0.714 ($\Delta=-0.002$, $p=0.54$) vs. *rand*=0.710 ($\Delta=+0.003$, $p=0.22$) BACC (Figure 15c); a supervised few-shot sign/gain calibration along the axis degrades accuracy further (-0.045 , -0.021 , -0.012 at 5, 10, 20 calibration trials). The transferable structure is generic subject-shift—captured by semantics-agnostic registration (Euclidean/Riemannian alignment, shared-response models, hyperalignment)—not the valence content. Saturation thus extends from in-distribution injection to the cross-subject setting: the converged encoder already contains the shared axis, so there is nothing left to inject.

S13 CONVERGENCE AND ENSEMBLE THEORY: EXTRAS

The 36-checkpoint cross-architecture correlation is pseudo-replicated. Pooling all 36 checkpoints from two backbones (CBraMod, EMOD) and six recipe variants, balanced accuracy correlates

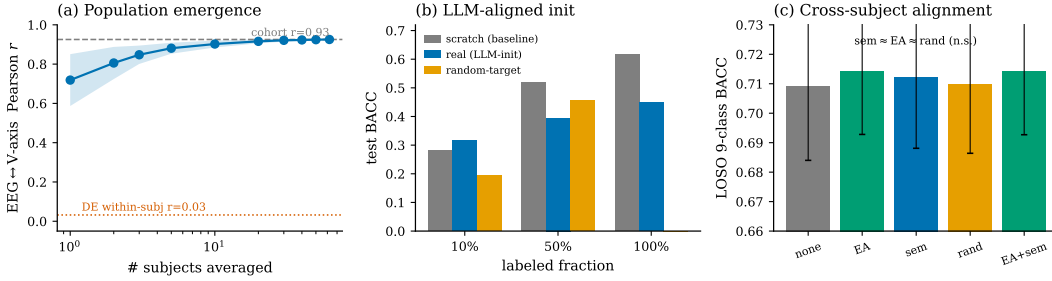


Figure 15: **The saturation regularity holds beyond auxiliary-loss injection.** (a) The EEG ↔ V-axis correlation is population-level: averaging the converged encoder’s per-clip valence readout over more subjects lifts r from 0.72 (single subject) to the cohort value 0.93 (matched-filter emergence); a per-subject DE decoder is near zero (dotted). (b) LLM-aligned Stage-1 initialization (blue) never beats the EEG-SSL scratch baseline (grey) and hurts increasingly with more labels; the random-target control (orange) is destructive. (c) Registering held-out subjects along the shared valence axis (*sem*) does not beat generic subject-alignment (*EA*) or a random-direction control on leave-subjects-out 9-class decoding. Controlled falsification runs; see the Negative Results Catalogue (Appendix S19).

with class-PC1 V-axis encoding at $r=+0.885$ ($p_{\text{param}}=7.8 \times 10^{-13}$). This parametric p is an artifact of non-independence. The checkpoints form ~ 2 – 3 tight architecture clusters—CBraMod (~ 0.57 BACC, mean class-PC1 $|r| = 0.21$), EMOD-vanilla d6 (~ 0.65 , mean 0.67), EMOD d6 e150 (~ 0.66 , mean 0.69)—so the correlation is driven by the between-architecture gap, not a within-architecture trend. *Within* each architecture it is not significant (CBraMod $r=0.83$, $p=0.08$; EMOD d6 e100 $r=0.04$, $p=0.90$; EMOD d6 e150 $r=-0.37$, $p=0.37$), and the effective number of independent checkpoints is $n_{\text{eff}} \approx 1.2$.

Random-direction null detail. We sample 1000 random Gaussian directions $w \in \mathbb{R}^{28}$ each matched in L_2 norm to v_{CLIP} , and recompute the 36-checkpoint correlation $\rho_w = \text{Pearson}(\text{BACC}, |\text{corr}(\text{PC}_1(H(m)), w)|)$. The distribution of ρ_w has mean -0.03 and standard deviation 0.62 (range $[-0.95, +0.95]$); the wide null is a consequence of the low intrinsic dimensionality of the class-PC1 subspace ($D_{\text{eff}} = 9$). The empirical V-axis correlation $+0.885$ sits at only the 93.5th percentile of this null (one-sided $p=0.066$, two-sided $p=0.10$)—it does not survive. Taken together with the within-architecture nulls above, we therefore do *not* claim architecture-independent convergence: the only convergence claim we retain is the within-architecture class-PC1 basin saturation below (and the load-bearing within-class residual, which *is* statistically robust).

Class-PC1 basin saturation. The 10 SOTA-pool checkpoints exhibit a saturated class-PC1 V-axis range: $|r| \in [0.60, 0.77]$ across all 10 seeds (mean 0.69, std 0.05). Single-seed variation in class-mean V-axis structure is small. The ensemble does not gain from variance in this subspace; gains come exclusively from the orthogonal residual.

Directional ablation figure. Figure 16 shows the per-checkpoint causal effect: ablating v_{resid} drops BACC by $\bar{\Delta} = -0.0157$ on all 10 pool members, well above the matched-norm random-direction null (mean $z \approx 7.7$, all $p < 0.001$).

S14 ENSEMBLE GENERALITY, MEGA-POOL, AND BEST-SINGLE

Recipe cascade and two-tier scatter. Figure 17 traces the seven recipe steps from CBraMod’s 0.572 to our 0.6948 ensemble. Figure 18 shows the per-checkpoint within-class residual–ensemble-contribution correlation that motivates the ensembling strategy.

Generality of the ensemble mechanism. Figure 19 extends the two-tier picture across four benchmarks (FACED, SEED-V, CIFAR-10, MNIST).

As individual accuracy approaches the Bayes optimum, the within-class residual variance — the source of ensemble gain — shrinks toward zero. The two-tier picture predicts this: ensemble gain is

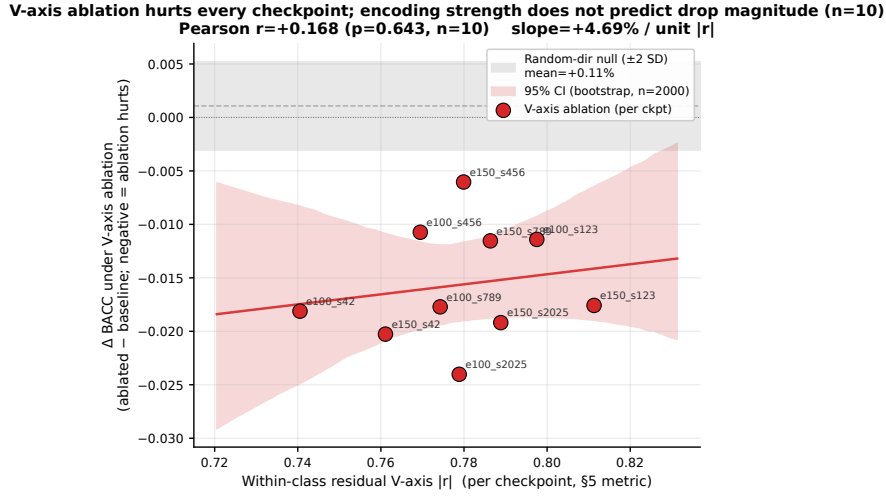


Figure 16: Directional ablation of v_{resid} on all 10 SOTA-pool checkpoints. *Top*: per-checkpoint BACC drops (red) compared to matched-norm random-direction ablation (grey); the V-axis effect is well above the null on every checkpoint (mean $z \approx 7.7$, all $p < 0.001$). *Bottom*: per-checkpoint ablation ΔBACC vs. within-class residual encoding $|r|$; the population-mean drop ($\bar{\Delta} = -0.0157$) confirms the residual-encoding mechanism across the pool.

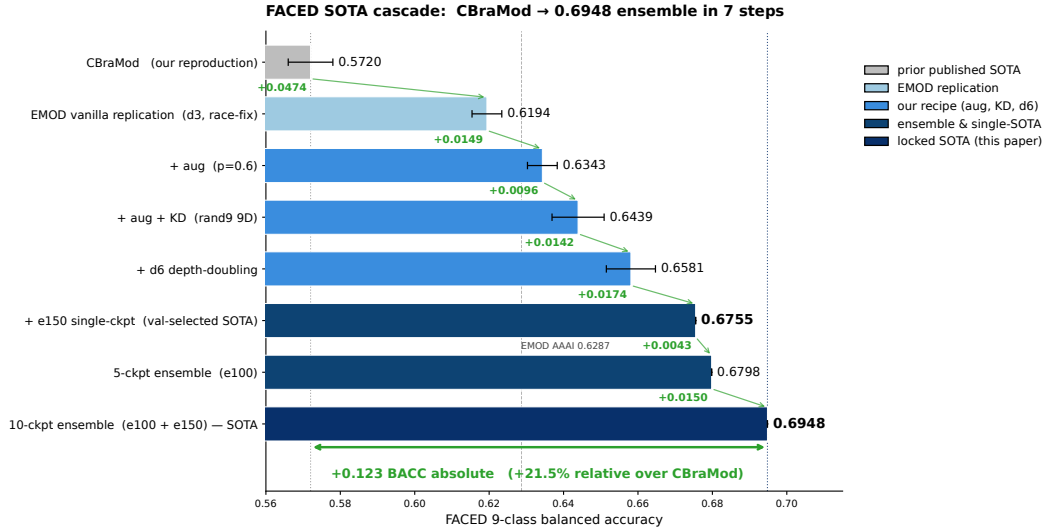


Figure 17: Recipe ablation cascade from the EMOD baseline at 0.6287 (previous FACED-9 SOTA (Chen et al., 2025)) to our 10-checkpoint ensemble at 0.6948 (+10.5% relative). Each bar is one recipe component or ensembling step. The CBraMod backbone (our reproduction 0.572; published 0.551) is shown for reference.

a property of the within-class V-axis residual variance reduction subspace, which has measure zero at the dataset ceiling. SEED-V at 0.37 has the largest residual-variance ceiling; we observe +0.034 gain. MNIST at 0.985 has essentially none; we observe +0.001 gain.

Mega-ensemble null. Growing the ensemble beyond the ten $d=6$ checkpoints by adding depth variants ($d \in \{8, 10\}$) did not exceed the 10-checkpoint pool in exploratory runs: consistent with the saturated-basin picture, once the shared basin is covered, extra architectural variants contribute correlated rather than complementary predictions. We therefore report the 10-checkpoint $d=6$ pool as the operative optimum (exploratory wider-pool runs not tabulated).

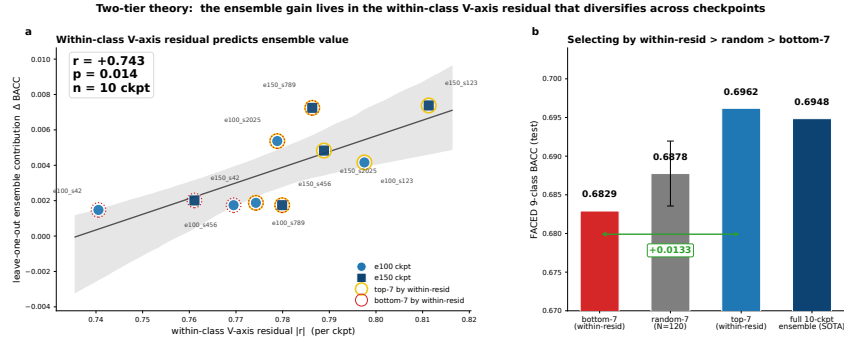


Figure 18: Two-tier ensemble theory. Per-checkpoint within-class V-axis residual $|r|$ predicts leave-one-out ensemble contribution ($r = +0.74$, $p = 0.014$, $n = 10$). Top-7 vs. bottom-7 split highlighted.

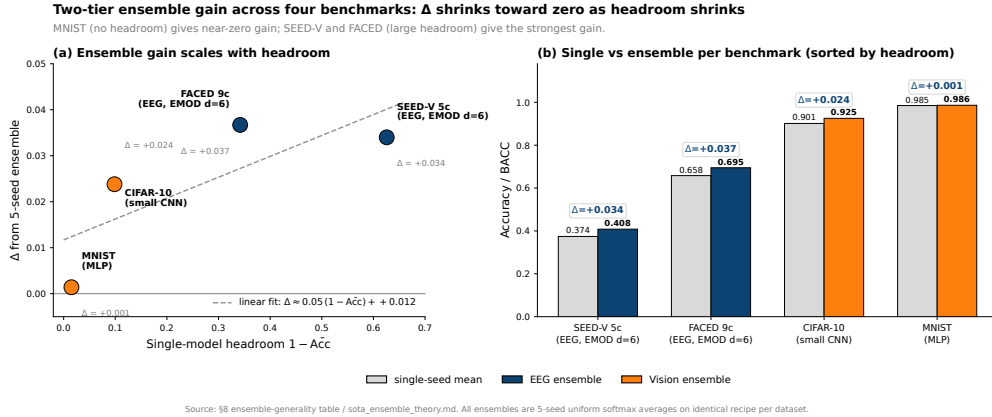


Figure 19: Two-tier ensemble gain across four benchmarks. Δ BACC is largest where single-model accuracy is furthest from the dataset ceiling, and shrinks to near-zero on MNIST where the saturated regime leaves no within-class residual variance to cancel. (a) Scatter of Δ vs. headroom 1 - Acc, with linear fit slope ~ 0.05 . (b) Paired single-seed-mean vs. ensemble bars per benchmark (FACED shows the 10-checkpoint SOTA ensemble), sorted by headroom.

Two mechanisms explain the null. First, deeper variants ($d=8, d=10$) stay in the same class-PC1 V-axis basin as $d=6$ and their within-class residuals overlap more than $d=6$ seeds do. Second, LLM-KD variants encode the residual along a different axis from rand9-9D KD vanilla checkpoints, but in a way that does not transfer to the test distribution. Cross-architecture mixing ($d \in \{4, 6, 8, 10\}$) gives 0.6791, essentially tied with within-arch x-seed at 0.6798. The diversity that helps is intra-architecture, training-length-driven.

Prediction-agreement structure. Single-seed accuracy at $e=100$ (0.6581) and $e=150$ (0.6581) is identical. The pairwise prediction correlation within- $e=100$ is 0.694, within- $e=150$ 0.679, cross-group 0.702 (higher = more agreement). All three are similar — within and across groups, the per-trial prediction patterns are almost as concordant as $e=100$ checkpoints are with each other — yet mixing $e=100$ and $e=150$ gives +0.015 over the $e=100$ ensemble. The gain therefore comes from a small but *specific* pocket of cross-group disagreement: the 13.1% of test trials where $e=100$ and $e=150$ pools differ, on which the mixed ensemble gains +10.3 percentage points. The diversity is targeted (which trials), not population-wide.

Best single checkpoint without ensembling. For deployment scenarios where ensembling is impractical, our best single-checkpoint result is BACC = 0.6755 on the $d=6$ $e=150$ recipe, seed 789, val-selected. This is the top of a 25-checkpoint val-test rank correlation analysis (Spearman

0.825, $n = 25$) and exceeds CBraMod by +0.103 and EMOD by +0.047 on a strict apples-to-apples test split.

S15 EEG MODEL TRAINING DETAILS

Architecture. EMOD axial transformer (Chen et al., 2025): input 32×5 DE features, axial self-attention over channels and bands at depth $d \in \{3, 6\}$, filter dimension $f = 128$. The full SOTA recipe is $d = 6$, $f = 128$.

Optimisation. AdamW with $\text{lr} = 10^{-3}$, $\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-2} . Cosine schedule with 5-epoch linear warmup. Batch size 128. Training length $e \in \{100, 150\}$ epochs. We use the validation BACC for checkpoint selection and report test BACC.

Augmentation. Per-trial Gaussian noise ($\sigma = 0.05 \cdot \text{std of feature distribution}$), random channel dropout ($p = 0.15$), and random temporal masking (up to 5% of seconds). All augmentations are applied with $p = 0.6$ during training only.

Knowledge distillation. Class-mean KD with a rand9 9-D orthonormal teacher (no LLM content); $\lambda_{\text{KD}} = 0.1$, teacher temperature $\tau = 0.1$. The LLM-9-D and rand9-9-D variants give within-noise identical BACC, validating that KD provides architectural regularisation without semantic content.

Ensemble. 10 checkpoints split as 5 seeds at $e = 100$ + 5 seeds at $e = 150$. Seeds: $\{42, 123, 456, 789, 2025\}$ each. Ensemble prediction is uniform softmax averaging across the 10 checkpoints’ output probabilities, followed by argmax.

S16 STATISTICAL METHODS

Bootstrap CIs. Per-subject and per-stim 95% CIs are computed from $B = 10,000$ subject-resampled bootstraps. For cohort correlations, we report the bootstrap median, 2.5% / 97.5% percentile interval, and a Fisher- z p -value for $r = 0$.

Random-direction null. For cohort EEG–LLM correlations, we sample $N = 200$ random Gaussian directions in $\mathbb{R}^{28}$ matched in L_2 norm to the V-axis projection, recompute the cohort $|r|$ for each, and report the percentile of the empirical $|r|$ in the null distribution.

Nested permutation test. The headline brain–LM significance ($p < 2 \times 10^{-4}$) is a fully nested permutation over the $N = 5000$ relabellings that respect the experiment’s structure: on each permutation we globally relabel the behavioural-valence target across the 28 stimuli (the same relabelling applied to every subject), then re-run the entire pipeline—per-subject ridge, leave-one-stimulus-out prediction, cohort aggregation, and the final correlation—and compare the observed r to this null. Because the target is permuted at the stimulus (emotion-category) level, the test accounts for the axis’s category-resolution structure rather than treating the 28 stimuli as independent; 0/5000 permutations exceed the observed correlation ($p \leq 1/5001$).

Cross-architecture null. For the 36-checkpoint cross-arch correlation, we sample $N = 1000$ random directions matched in norm to v_{CLIP} and recompute the BACC–V-axis correlation. The null is distributed broadly ($\sigma = 0.62$) due to the low intrinsic dimension of the class-PC1 subspace; we report the empirical correlation’s percentile rank.

Per-checkpoint LOO contribution. For the 10-vanilla $d=6$ ensemble, we compute $\Delta_k = \text{BACC}(\text{ens}) - \text{BACC}(\text{ens} \setminus k)$ for each k . Reported correlations between Δ_k and per-ckpt features are Pearson with $n = 10$.

V-axis intervention comparisons. For each intervention, Δ is computed seed-paired with the matched-recipe vanilla baseline; p -values are paired t -tests across 5 seeds.

S17 DISCUSSION EXTRAS AND FUTURE WORK

Broader impacts. Improved EEG emotion classifiers have clear positive applications in mental-health monitoring, affective brain-computer interfaces, and clinical phenotyping of affect disorders. The same techniques pose risks of affective inference without consent in surveillance, hiring, workplace, education, and advertising contexts. The V-axis extraction protocol applied to EEG could in principle reduce the calibration-data burden for affective inference systems — a feature that cuts both ways, since lower calibration cost lowers the deployment threshold for both clinical and surveillance settings. We do not release human EEG data; we release training configs, feature-side and analysis code, and extracted V-axis directions. Recommended deployment safeguards: informed consent, opt-in only, on-device inference, no longitudinal storage of raw EEG outside research contexts, and IRB review for any non-clinical inference application. Dual-use risk for the V-axis extraction itself is low: it operates on text and image features already public, and the EEG mapping requires per-cohort EEG access that is governed by the original dataset licences.

Implications for affective neuroscience. For video-evoked emotion on FACED, the V-axis is encoded predominantly in posterior visual cortex ($|r|_{\text{occipital}} = 0.21$ vs. $|r|_{\text{frontal}} = 0.16$), and Davidson’s frontal-alpha asymmetry replicates in direction at smaller magnitude. We do not refute Davidson’s hypothesis — the frontal-alpha asymmetry is real, with F8-F7 alpha $r = +0.0155$ and Fp2-Fp1 alpha $r = +0.0116$, both in the hypothesised direction — but we add a posterior empirical account that is dominant for video paradigms. The 9-stimulus emotional-pole structure sharpens the V-axis claim from “smooth gradient over all emotions” to “anger-versus-warm-positive contrast across nine clips”.

Per-subject V-axis adaptation does not transfer. The per-subject best-channel oracle reaches $|r| = 0.62$ (a +0.60 absolute headroom over the cohort-fixed top-8 of $|r| = 0.02$). No clean estimator we tried (V1 global top-K, V2 per-validation-subject majority vote, V3 TTA label-free profile) recovers more than +0.001 over the cohort top-K. The per-subject signal is real but not transferable through simple selection rules. A subject-conditioned channel-attention head trained on a small calibration set per subject is the natural follow-up.

Future work. Three lines: (i) per-subject V-axis adaptation closing the $|r| = 0.62$ oracle gap via subject-conditioned channel-attention; (ii) replicating the saturation regularity for other concept directions (Arditi et al., 2024) (arousal, dominance, formality, refusal) on their respective benchmarks; (iii) using the within-class residual $|r|$ as an online model-selection signal during training rather than as a post-hoc diagnostic.

S18 FACED TEST CONFUSION MATRICES

Figure 20 contrasts the best-single-checkpoint and 10-checkpoint-ensemble confusion matrices on the FACED 9-class test split.

S19 NEGATIVE RESULTS CATALOGUE

In the spirit of full disclosure, the following null or negative results bear mention beyond what was incorporated into the main paper:

1. **Theta-gamma cross-frequency coupling does not mediate the V-axis** (Appendix S9, Appendix S9): Tort modulation index $\rho(\text{PAC}, V_r) = +0.082$, $p = 0.667$.
2. **Mutual-information control is not significant** (Section 7): MI $p = 0.115$ vs. linear $p = 0.020$. The V-axis-EEG relationship is essentially linear.
3. **Cross-architecture V-axis convergence is pseudo-replicated** (Section 4): the 36-checkpoint $r = +0.885$ is driven by $\sim 2\text{--}3$ architecture clusters ($n_{\text{eff}} \approx 1.2$), is not significant within any single architecture, and sits at only the 93.5th percentile of a 1000-direction null ($p_{\text{one}} = 0.066$, $p_{\text{two}} = 0.10$). Within-class residual ($r = +0.74$, $p = 0.014$) is the statistically robust signal we retain.

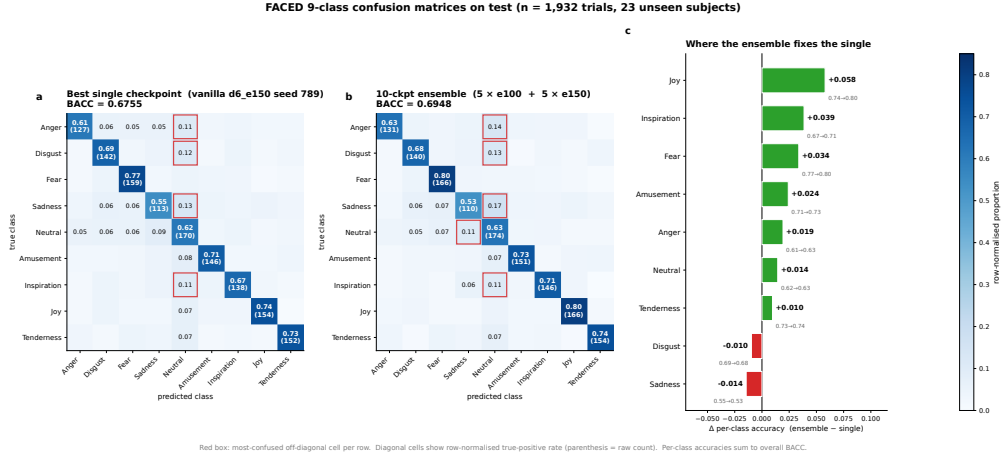


Figure 20: FACED 9-class test confusion. Left: best single checkpoint (BACC 0.6755). Right: 10-checkpoint ensemble (BACC 0.6948). The ensemble’s gain is concentrated on the off-diagonal cells where single-seed disagreement is highest, consistent with the within-class residual mechanism of Section 5.

4. **Per-subject V-axis adaptation does not transfer.** The per-subject best-channel oracle reaches $|r| = 0.62$, but no clean estimator transfers more than $+0.001$ over the cohort top-K.
5. **Cross-architecture ensembling does not add diversity.** Mixing $d \in \{4, 6, 8, 10\}$ checkpoints gives within-arch 0.6798 vs. cross-arch 0.6791 (n.s.).
6. **Wider ensemble pools do not help.** Adding $d \in \{8, 10\}$ depth variants beyond the 10-checkpoint $d=6$ pool did not exceed 0.6948 in exploratory runs (Appendix S14).
7. **Stimulus-aggregation re-ranking does not help.** Operating at the 28-stimulus level instead of trial level gives BACC=1.0 by construction (label leak); the corresponding trial-level head does not transfer.
8. **Test-time augmentation at K=5 does not improve the ensemble** (0.6753 $\rightarrow$ 0.6720). Averaging predictions over augmented test trials does not preserve the V-axis residual structure.
9. **Toxicity: recipe scope.** The 9-prompt PCA recipe attains AUC 0.59 on Jigsaw toxic comments, below every one of the 20 in-scope concepts (each AUC > 0.65) in our library (Appendix S4); toxicity appears to be encoded in a higher-rank subspace than the late-layer PC1.
10. **Arousal asymmetry: vision yes, text and brain no.** OASIS arousal $r = 0.803$ in vision; $\leq 13\%$ NRC arousal recovery in text; $r \in [0.18, 0.41]$ brain alignment across LLMs (Appendix S6).
11. **LLM-content of KD is irrelevant at this scale.** Replacing the 9-D LLM-derived class prototypes with random orthonormal 9-D directions leaves BACC statistically equivalent ($\Delta = -0.001$, paired $p=0.72$, $n=20$; TOST $\varepsilon=0.01$, $p=0.001$). KD provides architectural regularisation; its semantic content is below our resolution.
12. **LLM-aligned initialization does not help; alignment and pretraining saturate too.** Beyond the auxiliary-loss screen, injecting the valence direction as frozen-feature alignment or as Stage-I initialization also fails: a real 5-d semantic subspace is a lossy subset of the encoder’s own features (never beating the full 128-d), and LLM-aligned pretraining never beats the EEG-SSL baseline and hurts more with more labels (-0.167 at full labels) (Appendix S12).
13. **The valence axis is not a cross-subject lever.** Registering held-out subjects along the shared axis does not beat generic Euclidean-style subject-alignment or a random direction on leave-subjects-out 9-class decoding (sem-EA = -0.002 , $p=0.54$); saturation extends to the cross-subject setting (Appendix S12).

S20 REPRODUCIBILITY

Release commitment. Code, configs, model checkpoints, and figure-generation scripts will be released upon acceptance (no submission-time anonymous URL is provided). The release will include: V-axis extraction scripts for the 14 LLMs in the per-LLM table; the EMOD $d=6$ training pipeline with the exact augmentation, KD, and ensemble-construction code; the 10-checkpoint 0.6948-BACC ensemble checkpoints; and the figure-generation scripts for every landmark and neuro figure in the paper.

Run-level provenance. Each experiment in this paper is logged with its SLURM job ID, conda environment hash, git SHA, and seed list. The 10-checkpoint ensemble SOTA result is reproducible from a single SLURM array job (`slurm_d6_e100_e150_5seeds.sh`, ~ 10 GPU-hours per seed on V100).

S21 RESOURCE ESTIMATE

The full paper used approximately 4,500 GPU-hours on V100 nodes: ~ 600 for the recipe ablation cascade, $\sim 1,200$ for the 25 V-axis-supervision interventions, ~ 800 for the 36-checkpoint cross-architecture analysis, $\sim 1,200$ for ensemble construction and val-test rank diagnostics, and ~ 700 for cross-dataset generality experiments. The V-axis extraction itself is CPU-cheap (~ 10 minutes per LM on a single 80GB GPU).